



CERTIFICATE OF CONFORMANCE

QTR WO#: Q34656-1

Customer: Energy Equipment

PO#: 250504-2

Description: 10" 900# RFFE Fig 55 Delta Ball Valve, G/O

Quantity: 3

S/N(s): R2217500040001, R2226600040001,
R2226600040002

THE EQUIPMENT LISTED ABOVE CONFORMS TO THE FOLLOWING:

B16.34: ☒

NACE MR-0175; ☒
Part 1 & 2:

Fire-Safe per ☒
API 6FA:

Pressure Balance / Communication Hole:

Yes: ☐ No: ☒

API 6D 23rd Ed: ☐

API 6D 24th Ed: ☒

API 6D 25th Ed: ☒

We certify that the material listed above has been manufactured and tested in accordance with the current revisions listed above (edition, errata, and addenda), Valvitalia standards, and customer purchase order requirements. Material listed has successfully passed all mandatory test and meets supplemental requirements as noted above.

Michael Carson
Quality Assurance Manager

Date: 7/16/25

Body: R2217500040001

 forge nicromo SpA STABILIMENTO/PLANT: VIA PROVINCIALE, 57 - 23813 BINDO DI CORTENOVA (LC) - ITALY TEL: 0039-0341-979831 E-MAIL: info@forgenicromo.it WEB: www.forgenicromo.it Sede legale/Reg. office: Via P.R. Giuliani 10, c/o Studiotesta - 20900 Monza (MB) - Italy Sede amministrativa/Head office: Via Provinciale, 57 - 23813 Bindo di Cortenova (LC) - Italy Capitale sociale/Share capital € 1.000.000,00 - Partita IVA/VAT n° 00901140962 Cod. fiscale/Reg. imprese Monza Brianza (MB) 08287210150 - REA MB-1216530		CERTIFICATO PROVE MATERIALI N° TEST CERTIFICATE EN 10204 / 3.1		2201641 Sh 1												
		DATA/DATE/ DATE/DATUM 15/11/2022		COMMESSA/JOB/ COMMANDE/KOMM 2201641												
		ORDINE/AUFTRANG/ ORDER/COMMANDE OF000065954														
		CUSTOMER/CLIENT/KUNDE VALVITALIA SpA Via del Lauro, 7 20121 MILANO (MI)														
	COLATA heat coulé schmelze	CERT. ORIGINE mill certificate certificat d'origine werkszeugnis	NORMA MATERIALE material grade norme material werkstoff	DESCRIZIONE description description gegestand	QUANTITA quantity quantité stückzahl	ITEM										
	289476	ACCIAIERIE VENETE SpA	ASTM A350 LF2	BALL VALVE BODY 10" #900 V000043880 7UCP109U10 Progetto R22175U10 20	10	2										
ANALISI CHIMICA/ chemical analysis / analyse chimique / chemische Zusammensetzung																
	COLATA heat coulé schmelze	C %	Mn %	Si %	P %	S %	Cr %	Ni %	Mo %	Cu %	V %	Al %	Nb %	Ti %	N %	CE %
	289476	0.23 max.	0.6 - 1.35	0.15 - 0.3	0.025 max.	0.02 max.	0.3 max.	0.4 max.	0.12 max.	0.4 max.	0.08 max.		0.02 max.			0.42 max.
	289476	0.161	1.123	0.235	0.01	0.005	0.12	0.121	0.033	0.199	0.004	0.028	0.003	0.008	0.0078	0.40
CARATTERISTICHE MECCANICHE/ mechanical tests according to ASTM A370:																
	COLATA heat coulé schmelze	SNERVAM. Yield Strength MPa 275 min.	ROTTURA Tensile Strength MPa 485 - 655	ALLUNG. Elongation % 22 min.	STRIZIONE Reduction of Area % 30 min.	°C temp	HARDNESS HB 197 max.	IMPACT TEST KV 10X10X55 mm		°C temp						
	289476	300	493	33	71	+20	146 146 144	SINGLE 21 J min.		AVERAGE 27 J min.						
HEAT TREATMENT: NORMALIZED AT 900°C FOR 2h, COOLING IN AIR		MATERIAL IN ACCORDANCE WITH: NACE MR01-75 / ISO 15156 ASTM A350 LAST ED. / MDS 001 Rev.3 / QTR STOCK - 7U Rev.8 Qualified material manufacturer according 2014/68/UE Annex I Section 4.3 (EN 764-5:2014)			NOTES: Reduction ratio > 4:1											
CE = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15		STEEL MAKING PROCESS: ELECTRIC FURNACE, VACUUM DEGASSED, FULLY KILLED FINE GRAINED														
VISUAL INSPECTIONS AND DIMENSIONAL CHECK : SATISFACTORY				MATERIAL FREE FROM RADIOACTIVE CONTAMINATION												
DATE: 15/11/2022		ISSUED BY: Q.A.D. S. RAPEZZI		APPROVED BY: Quality Mgr S. CERLETTI		T.P.I.										

THE MATERIAL DESCRIBED ABOVE IS IN COMPLIANCE WITH ORDER'S REQUIREMENTS



 SIDERFORGEROSS 36011 Arsiero (VI) ITALY - Via Cartiera di Mezzo, 38 Tel. 0445 714400 - Fax 0445 714444	Inspection Certificate Quality Control Department		COMPANY WITH QUALITY SYSTEM CERTIFIED BY DNV = ISO 9001 & EN 9100 = COGOLLO PLANT 36010 Cogollo del Cengio (VI) ITALY - Via M. Schiro, 2
	N° 9808 / 22 REV.0 Date 29/11/2022		
Purchaser: VALVITALIA S.P.A. PIAZZA SIGMUND FREUD 1 20154 MILANO MI	Order No.: OF000066001	Certificate type: EN 10204 ed. MARCH 2005 ☐ 2.2. ☒ 3.1. ☐ 3.2	
	SFRG job No.: OV223715	Melting practice ☒ E.F.V.D ☐ E.F ☐ AOD ☐ VOD ☐ EB ☐ VAR ☐ VIM ☐ ESR	

CHEMICAL ANALYSIS / LADLE ANALYSIS

MATERIAL: ASTM-A350-LF2-CL.1 ED.21

HEAT: 135282

SUPPLIER: Pallanzeno s.r.l.

C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	Sn	V	Nb	N	Pb	H	B	Ca		
%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	ppm	%		
0,16	1,16	0,26	0,13	0,1	0,002	0,013	0,033	0,22	0,03	0,005	0,009	0,003	0,001	0,006	0,001	0,9	4	0,0018		

C.E. (Long) = 0,400

HEAT TREATMENT

TYPE	TEMP °C	HOURS	COOLING	TYPE	TEMP °C	HOURS	COOLING
NORMALIZING	900	5,5	AIR				

MECHANICAL TESTS

		MPa	MPa	MPa	%	%	Dir.	Type	Temp °C	Avr/Min	J	Min	Max	
FROM PIECE	☐	MIN	485	275		22	30	TAN	KV	-46	27	21	148	197
FROM QTC	☒	MAX	655											

Treated with the Pcs

OBTAINED

N° TEST	DIM	SECT.	LENGHT	TEMP	DIR	UTS	YS 0,2	YS	EL	RA	IMPACT TEST								HB on pieces	
	mm	mm	mm	°C		MPa	MPa	MPa	%	%	Dir.	Type	Temp °C	JOULE				Avr	Min	Max
6104	12,50	122,72	50,00	+20	TAN	496	311		33,5	75,0	TAN	KV	-46	52	62	60	58		150	150

OUR ITEM	CUSTOMER ITEM	DESCRIPTION	DRAWING	QUANTITY
7	from 70 to 70	CLOSURE TYPE 7U GREZ. 10" #900 546x254x291	V000043783	20
	from to			
	from to			
	from to			
	from to			
	from to			

REMARKS:

PIECES CONFORMING TO DIMENSIONAL AND VISUAL CONTROL. REDUCTION RATIO > 3:1
 MATERIAL ACCORDING TO MDS_001 Rev. 3, ITP: QTR STOCK - 7U Rev.8
 MATERIAL ACCORDING TO NACE MR 01-75/ISO 15156 ed. 2015.
 QUALITY ASSURANCE SYSTEM IN ACCORDANCE WITH THE P.E.D. 2014/68/EU ANNEX 1 PARA. 4.3.
 HEAT TREATMENT FURNACE CALIBRATED ACCORDING TO API 6D 25th ED. ANNEX H.
 REF. S.N. FROM 1 TO 20



17 APR. 2023

Note: Prodotti conformi ai requisiti dell'ordine - Products are in compliance with the requirements of the order - Les produits sont conforme à votre ordre - Produkt entsprechend bestellungs spezifikation.

Material conforming ☒	Quality Control Dept. 	Surveyor:
Material non conforming ☐		

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SIDERFORGEROSSİ
GROUP

 36011 Arsiero (VI) ITALY - Via Cartiera di Mezzo, 38
 Tel. 0445 714400 - Fax 0445 714444

Inspection Certificate

Quality Control Department

N° 8671 / 22 REV.0 Date 02/11/2022

 COMPANY WITH
 QUALITY SYSTEM
 CERTIFIED BY DNV
 = ISO 9001 & EN 9100 =

Purchaser:

 VALVITALIA S.P.A.
 PIAZZA SIGMUND FREUD 1
 20154 MILANO MI

Order No.:

OF000066001

Certificate type: EN 10204 ed. MARCH 2005

☐ 2.2. ☐ 3.1. ☐ 3.2

SFRG job No.:

OV223715

 Melting ☐ E.F.V.D ☒ E.F. ☐ AOD ☐ VOD
 practice ☐ EB ☐ VAR ☐ VIM ☐ ESR

CHEMICAL ANALYSIS / LADLE ANALYSIS

MATERIAL: ASTM-A350-LF2-CL.1

HEAT: 283963

SUPPLIER: ACCIAIERIE VENETE S.p.A.

C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	Sn	V	Nb	N	Pb	As	H	B	Ca	
%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	
0,15	1,119	0,229	0,13	0,165	0,007	0,01	0,029	0,163	0,045	0,008	0,008	0,007	0,001	0,0059	0,0004	0,005	1,4	0,0002	0,0011	

C.E. (Long) = 0,399

HEAT TREATMENT

TYPE	TEMP °C	HOURS	COOLING	TYPE	TEMP °C	HOURS	COOLING

MECHANICAL TESTS

REQUIRED	UTS	YS 0,2	YS	EL	RA	IMPACT TEST					HB on pieces	
	MPa	MPa	MPa	%	%	Dir.	Type	Temp °C	Avr/Min	J	Min	Max
MIN	485	275		22	30	LNG	KV	-46	27	21		197
MAX	655											

Treated with the Pcs

OBTAINED

N° TEST	DIM	SECT.	LENGHT	TEMP	DIR	UTS	YS 0,2	YS	EL	RA	IMPACT TEST								HB on pieces	
	mm	mm	mm	°C		MPa	MPa	MPa	%	%	Dir.	Type	Temp °C	JOULE	Avr	Min	Max			
3597	12,50	122,72	50,00	+20	LNG	503	336		34,5	73,0	LNG	KV	-46	56 67 64 62		148	152			

OUR ITEM	CUSTOMER ITEM	DESCRIPTION	DRAWING	QUANTITY
6	from 60 to 60	SFERA VALVOLA SFERA TYPE 5U GREZ. 10" #150 - 387x254x288 F	5USF101U00..GRev0	V000042485
	from			10
	from			
	from			
	from			
	from			
	from			

REMARKS:

PIECES CONFORMING TO DIMENSIONAL AND VISUAL CONTROL.

MATERIAL ACCORDING TO SPEC.: MDS_001 REV. 3 AND ITP QTR STOCK - 7U REV. 8.

MATERIAL ACCORDING TO NACE MR 01-75/ISO 15156 ed. 2015. REDUCTION RATIO > 3:1.

QUALITY ASSURANCE SYSTEM IN ACCORDANCE WITH THE P.E.D. 2014/68/EU ANNEX 1 PARA. 4.3.

FOR CHEMICAL ANALYSIS SEE FORGEROSSİ RESEARCH REPORT No L221705/A.

HEAT TREATMENT FURNACE CALIBRATED ACCORDING TO ANNEX M API 6A 21st Ed. AND ANNEX H API 6D 25th Ed.



17 APR. 2023

Note: Prodotti conformi ai requisiti dell'ordine - Products are in compliance with the requirements of the order - Les produits sont conforme à votre ordre - Produkt entsprechend bestellungs spezifikation.

 Material conforming ☒

 Material non conforming ☐

Quality Control Dept.:

 SIDERFORGEROSSİ
 Q.C. FENOGLIO EMILIO

Surveyor:

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Stem: R2217500040001

DC0202 - Certificato di prodotto Rev. 2



CERTIFICATO : OR22 - 562/4
CERTIFICATE
EN 10204 3.1



Cliente : VALVITALIA S.p.A. Customer	Consegna : DDT 220769 of 30.11.2022 Delivery
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Vs Ordine : OF000066259 of 18.10.2022 Your order Pos. 320 - COD. V000146704 - COMM. R221757U10	Ns Ordine : 220426 of 26.10.2022 Our order
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Descrizione materiale : ROUND BAR PEELED AISI 4140 / 42CrMo4 HARDENED Material description NACE MR0175/ISO 15156 UNS G41400

Grado : 4140 Grade	Diametro : 110 mm. Diameter	Colata : T90995 Heat	Quantità : kg. 185,00 Quantity
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ANALISI DI COLATA / HEAT ANALYSIS

C	Mn	Si	P	S	Cr	Ni	Mo	Al	Cu	Sn	Ti	B	V	Nb	N	H
.417	.835	.273	.013	.002	1.010	.105	.191	.032	.151	.026	.004	.0001	.003	.002	.0076	1.4 ppm

PROVE MECCANICHE / MECHANICAL TESTS

Prove longitudinali eseguite +20°C alle condizioni di fornitura
 Longitudinal testing at +20°C in delivery condition

Snervamento Rp 0,2% : 658 N/mm ² Yield strength	Rottura Rm : 922 N/mm ² Tensile strength
Allungamento A5D % : 14 Elongation	Strizione Z% : 55 Striction
Resilienza KV -46°C : 63 - 72 - 58 J (Average 64.3 J) Impact value see report L.P.M. 22RP00530 of 04-02-2022	Durezza : 213 HB Hardness
Controllo spettrografico : N.D. Spectrographic examination	Ultrasuoni : N.D. Ultrasonic test
Dimensione grano : 8 Grain size	Corrosione : N.D. Corrosion Test
Trattamento termico : Quenching 860°C/4h/water and Heat treatment tempering 680°C/8h/air	Controllo visivo e dimensionale : satisfactory Visual and dimensional examination
Responsabile Qualita' : Quality control	Data : 30.11.2022 Date



09 GEN. 2023

ORIC ITALIANA S.r.l. Via dell' Industria, 4 Zona Ind.le 29015 Castel San Giovanni - PC - Italy
 tel. +39 0523 882 498 fax +39 0523 882 111
 www.oric.it e-mail info@oric.it

Seats: R2217500040001

 IRE- OMBA S.p.a ISO 9001:2015 ISO 14001:2015 ISO 45001:2018	Test certificate / Certificato Prove Materiale 3.1 EN 10204:2004	
	IRE1221156/5 	N° - Rev 1221156/5
Customer/Ciente: BBD S.R.L.		

Order/Ordine: **2022 AF 000645**

Inspection/Collaudo: IRE OMBA

Job/Comm.: **1221156**

Mater./Spec.:

Forging ratio / Rapporto di Fucinatura >= 4:1

Manufactur. process: forged+rolled rings

Manufactur. process: Forged + rolled rings

LF2 CL1 ASTM A350/18

Prova n° Test-Batch	Item Pos.	Pieces n° Pezzi	ø est mm	ø int mm	h mm	Identification/Identificazione	HBW1	HBW2
153468	010	10	320	275	68	G-AMA500-0009 2022 AF 000645	145	
153468	080	37	339	305	56	G-ANA500-1274 2022 AF 000645	148	145
153468	100	10	320	275	68	G-AMA500-0009 2022 AF 000645	145	
153468	110	8	320	275	68	G-AMA500-0009 2022 AF 000645	145	
153468	300	20	289	254	55	G-ANA500-1120 2022 AF 000645	145	148
153468	310	20	289	254	55	G-ANA500-1120 2022 AF 000645	145	148
153468	340	40	344	305	56	G-ANA500-1153 2022 AF 000645	145	148
153468	400	30	339	305	56	G-ANA500-1274 2022 AF 000645	145	148

Chemical analysis % / Analisi chimica

Heat n° / Colata			Steel mill / Acciaieria							
292825			ACC. VENETE S.P.A. - ITALY-(IT)							
Cert. orig. 615465			Vacuum degassed							
C	Mn	Si	P	S	Cu	Cr	Ni	Mo	Al	
0.1630	1.1570	0.2120	0.0100	0.0070	0.1100	0.0930	0.1120	0.0300	0.0220	
V	Nb	Ti	Sn	B	H	N	Ceq	P+S	Nb+V+Ti	
0.0050	0.0010	0.0080	0.0060	0.0003	0.00016	0.0058	0.396	0.017	0.014	
Cu+6Sn	Cr+Mo	CuCrMo	CuNiCrMo	CuNiCrMoV	NiCrMo	Nb+V				
0.146	0.123	0.233	0.345	0.35	0.235	0.006				

Heat treatment / Trattamento termico

Normalized 900°C 1.5h Air
 Furnace N°:TTS_F1
 Due date: 15/03/2023
 Furnace Calibration Doc. n°:22-0106/006

Tension Test/Trazione +20°C ASTM A370/ ASTM E8 $\sigma=12.5\text{mm}$ $L_0=50\text{mm}$

		Required values/Valori richiesti											
		Rp .2% N/mm2		Rp 1% N/mm2		Rm N/mm2		A4d %		A5d %		Z %	
		min	max	min	max	min	max	min	max	min	max	min	max
Sense/Senso		min	max	min	max	min	max	min	max	min	max	min	max
TANG		275				485	655	22				30	
		Obtained values/Valori rilevati											
Test n°	Sense/Senso	Rp .2% N/mm2		Rp 1% N/mm2		Rm N/mm2		A4d %		A5d %		Z %	
153468	TANG	312				516		32.5				67.4	

Impact test/Resilienza ASTM A370/ ASTM E23 10x10mm

Required values/Valori richiesti						Obtained values/Valori rilevati				
Sense/Senso	Type/tipo	T°C	Med./Av.	Min.	U.M	Test/Prova	Val1	Val2	Val3	Med./Av.
TANG	KV	-46	27	21	J	153468	85	90	82	85.67

Hardness test / Durezza: ASTM E10

Required values/Valori richiesti			Obtained values/Valori rilevati		
Min	Max	Type/tipo	Test/Prova	Val1	Val2
	197	HBW	153468	145	150

BBD S.r.l.
 Controllo Qualità
 13/04/2023



14 MAR. 2023

Data / Date (d/m/y) 12/01/23	Q.C DEPT. <i>Laura Gamba</i> GAMBA LAURA	Pag. 1 di 2
IRE - OMBA Spa Via Carlo Serassi, 14 24100 BERGAMO ITALY BG Te.: 035 / 41 60 811 Fax.: 035 / 41 60 856		

 IRE- OMBA S.p.a ISO 9001:2015 ISO 14001:2015 ISO 45001:2018	Test certificate / Certificato Prove Materiale 3.1 EN 10204:2004	
	IRE12211156/5 	N° - Rev 12211156/5
Customer/Ciente: BBD S.R.L.		

Order/Ordine: **2022 AF 000645**

Inspection/Collaudo: IRE OMBA

Job/Comm.: **12211156**

Spec./Note

MDS_001 Rev.3

API 6D 25th Edition Annex H

NACE MR0175/ISO15156-2 Ed.15

IR22099-01 R.4

Note



14 MAR. 2023

Data / Date (d/m/y) 12/01/23		Q.C DEPT. <i>Laura Gamba</i>  GAMBA LAURA	Pag. 2 di 2
IRE - OMBA Spa Via Carlo Serassi, 14 24100 BERGAMO ITALY BG Te.: 035 / 41 60 811 Fax.: 035 / 41 60 856			

Top Cover: R2217500040001

DC0202 - Certificato di prodotto Rev. 2



CERTIFICATO : OR22 - 6322/19
CERTIFICATE
EN 10204 3.1



Cliente : VALVITALIA S.p.A. Customer	Consegna : DDT 220769 of 30.11.2022 Delivery
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Vs Ordine : OF000066259 of 18.10.2022 Your order Pos. 310 - COD. V000138473 - COMM. R221757U10	Ns Ordine : 220426 of 26.10.2022 Our order
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Descrizione materiale : ROUND BAR FORGED LF2 CL.1 Material description	ASTM A350 - ISO 15156/NACE MR0175
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Grado : LF2 Grade	Diametro : 170 mm. Diameter	Colata : 202593 Heat	Quantità : kg. 108,00 Quantity
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ANALISI DI COLATA / HEAT ANALYSIS

C	Si	Mn	P	S	Cr	Ni	Mo	Al	Cu	V	Nb	Sn	Ti	H (ppm)	Ceq
.165	.280	1.040	.008	.002	.140	.120	.030	.030	.160	.003	.002	.011	.001	1.500	.392

PROVE MECCANICHE / MECHANICAL TESTS

Prove longitudinali eseguite +20°C alle condizioni di fornitura
 Longitudinal testing at +20°C in delivery condition

Snervamento Rp 0,2% : 334 N/mm2 Yield strength	Rottura Rm : 515 N/mm2 Tensile strength
Allungamento A % : 33 Elongation	Strizione Z% : 55 Striction
Resilienza KV -46 °C : 81 - 77 - 95 J (average 84.3 J) Impact value	Durezza : 152 - 156 HB Hardness
Controllo spettrografico : N.D. Spectrographic examination	Ultrasuoni : N.D. Ultrasonic test
Dimensione grano : 7 ASTM E112 Grain size	Rapporto di riduzione : 9 Reduction ratio
Trattamento termico : normalized 900°C/6h/air Heat treatment	Controllo visivo e dimensionale : satisfactory Visual and dimensional examination
Responsabile Qualità' : Quality control	Data : 30.11.2022 Date



19 DIC. 2022

ORIC ITALIANA S.r.l. Via dell' Industria, 4 Zona Ind.le 29015 Castel San Giovanni - PC - Italy
 tel. +39 0523 882 498 fax +39 0523 882 111
 www.oric.it e-mail info@oric.it

INSPECTION CERTIFICATE				CERT. TYPE		CMTR Nr.	SHEET			
LIST OF SUPPLIED PRODUCTS				ISO 10474 EN 10204		2203115/3 DATE 20/12/2022	1/6			
CUSTOMER : VALVITALIA S.P.A.				ORDER No.: OF000065821 Date: 30/09/2022						
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY				PROJECT No.:						
				JOB No.:						
LOT	OME ITEM	P.O. ITEM	Q.TY	Dimension	Drawing No.	Spec / Grade	Heat No.	Marking	Cond.	Coat.
L001	20	20	1152 No.	1.3/4" 8 UN	ASME B18.2.2-T.1.1.1-11 Ed.22	ASTM A 194 / A 194M API 20E-1 GR.7M	16/78540	OME.HEX-IT 7ML 20E1 351 A14	QTM	
224215	30	30	320 No.	1.3/4" 8 UNR x 210	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	211565	OME.IT-L7M 20E1- 352	QTM	
224230	30	30	320 No.	1.3/4" 8 UNR x 210	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	211565	OME.IT-L7M 20E1- 352	QTM	
224233	30	30	320 No.	1.3/4" 8 UNR x 210	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	211565	OME.IT-L7M 20E1- 352	QTM	
L004	40	40	170 No.	1.3/4" 8 UNR x 230	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	211565	OME.IT-L7M 20E1- 353	QTM	
L005	40	40	22 No.	1.3/4" 8 UNR x 230	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	BD4395	OME.IT-L7M 20E1- 353	QTM	
L006	140	140	1120 No.	1.1/8" 8 UN	ASME B18.2.2-T.1.1.1-11 Ed.22	ASTM A 194 / A 194M API 20E-1 GR.7M	18/70440	OME.IT-D 7ML 20E1 363 117	QTM	
L007	150	150	480 No.	1" 8 UNC	ASME B18.2.2-T.1.1.1-11 Ed.22	ASTM A 194 / A 194M API 20E-1 GR.7M	AJ3912	OME.IT 7LM 7ML 20E1 364 A8 HEX	QTM	
L008	160	160	480 No.	M20 x 2.5	UNI 5587	ASTM A 194 / A 194M API 20E-1 GR.7M	3082367	OME-IT 7ML 365 187	QTM	
L009	180	180	160 No.	1" 8 UNRC x 145	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	182664	OME.IT L7M 20E1 367	QTM	
L009	190	190	320 No.	1" 8 UNRC x 135	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	182664	OME.IT-L7M 20E1- 368	QTM	
L010	200	200	480 No.	M20 x 2,5 x 90	UNI 6610	ASTM A 320 / A 320M API 20E-1 L7M	212046	OME.IT-L7M- 369	QTM	
L011	210	210	254 No.	7/8" 9 UNC	ASME B18.2.2-T.1.1.1-11 Ed.22	ASTM A 194 / A 194M API 20E-1 GR.7M	1B5007602	OME.IT-D 7ML 370 356	QTM	
L011	210	210	66 No.	7/8" 9 UNC	ASME B18.2.2-T.1.1.1-11 Ed.22	ASTM A 194 / A 194M API 20E-1 GR.7M	1B5007602	OME.IT-D 7ML 370 356	QTM	
L012	220	220	800 No.	1.1/8" 8 UNR x 155	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	182916	OME.IT-L7M 20E1- 371	QTM	
L012	230	230	320 No.	1.1/8" 8 UNR x 165	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	182916	OME.IT-L7M 20E1- 372	QTM	
L013	260	260	100 No.	7/8" 9 UNRC x 125	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	182916	OME.IT-L7M- 375	QTM	
L013	280	280	220 No.	7/8" 9 UNRC x 115	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	182916	OME.IT-L7M- 377	QTM	
 03/02/2023										
VISUAL / DIMENSIONAL EXAMINATION : Conforming						MELT PROCESS : Electric Furnace				
OME Metallurgica Erbese S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it				Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION		Prepared 		Approved  F 8.2 - CMTR - 20		

INSPECTION CERTIFICATE										CERT. TYPE				CMTR N.				SHEET							
HEAT ANALYSIS										ISO 10474				2203115/3				2/6							
										EN 10204				3.1				DATE							
CUSTOMER : VALVITALIA S.P.A.										ORDER No.: OF000065821 Date: 30/09/2022															
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY										PROJECT No.:															
										JOB No.:															
LOT	Heat No.	C	Mn	Si	P	S	Cr	Ni	Mo	V	Ti	Cu	W	Al	B	Nb	Co								
		Zn	Pb	Sn	Be	O	N	H	Fe	Nb+Ta	Cb	Al+Ti	Cb+Ta	Se	Bi	Ca	Mg								
224215	211565	0,400	0,880	0,340	0,012	0,002	1,030		0,210																
224230	211565	0,400	0,880	0,340	0,012	0,002	1,030		0,210																
224233	211565	0,400	0,880	0,340	0,012	0,002	1,030		0,210																
L001	16/78540	0,395	0,830	0,180	0,025	0,022	0,910		0,190																
L004	211565	0,400	0,880	0,340	0,012	0,002	1,030		0,210																
L005	BD4395	0,410	0,850	0,190	0,013	0,005	1,020		0,170																
L006	18/70440	0,380	0,820	0,250	0,021	0,015	1,000		0,190																
L007	AJ3912	0,410	0,760	0,230	0,009	0,003	1,030		0,230																
L008	3082367	0,390	0,810	0,189	0,009	0,005	0,968		0,218																
L009	182664	0,400	0,770	0,230	0,008	0,003	0,980		0,200																
L010	212046	0,390	0,870	0,330	0,010	0,001	1,000		0,221																
L011	1B5007602	0,396	0,818	0,184	0,011	0,006	0,977		0,188																
L012	182916	0,400	0,800	0,240	0,009	0,004	0,990		0,200																
L013	182916	0,400	0,800	0,240	0,009	0,004	0,990		0,200																
 03/02/2023 OME Metallurgia Erbesse S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION Prepared  Approved  F 8.2 - CMTR - 29																									

INSPECTION CERTIFICATE							CERT. TYPE			CMTR Nr.			SHEET			
TENSILE TEST							ISO 10474 EN 10204 3.1			2203115/3 DATE 20/12/2022			3/6			
CUSTOMER : VALVITALIA S.P.A.							ORDER No.: OF000065821 Date: 30/09/2022									
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY							PROJECT No.:									
							JOB No.:									
LOT	SPECIMEN					REQUIRED VALUES							OBTAINED VALUES			
	NO.	DIA. mm	AREA mm2	L mm	T °C	Measure Unit	MIN Rp 0.2	R MIN MAX		MIN E %	MIN R.A. %	Rp 0.2	R	E %	R.A. %	
224215		12,5	122,66	50,00	Room	MPa	550	690		18	50	618	723	23,1	65,2	
224230		12,5	122,66	50,00	Room	MPa	550	690		18	50	621	730	24,0	63,4	
224233		12,5	122,66	50,00	Room	MPa	550	690		18	50	615	727	23,7	64,1	
L004		12,5	122,66	50,00	Room	MPa	550	690		18	50	620	725	23,9	63,2	
L005		12,5	122,66	50,00	Room	MPa	550	690		18	50	612	722	22,0	65,1	
L009		12,5	122,66	50,00	Room	MPa	550	690		18	50	616	718	23,2	62,7	
L010		12,5	122,66	50,00	Room	MPa	550	690		18	50	610	715	24,3	63,5	
L012		12,5	122,66	50,00	Room	MPa	550	690		18	50	607	721	25,1	64,0	
L012		12,5	122,66	50,00	Room	MPa	550	690		18	50	614	729	25,6	64,3	
L013		12,5	122,66	50,00	Room	MPa	550	690		18	50	612	723	23,4	65,8	
OME Metallurgica Erbe S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION Prepared  Approved  F 8.2 - CMTR - 05 03/02/2023 																

INSPECTION CERTIFICATE										CERT. TYPE				CMTR Nr.				SHEET							
IMPACT TEST										ISO 10474 EN 10204 3.1				2203115/3 DATE 20/12/2022				4/6							
CUSTOMER : VALVITALIA S.P.A.										ORDER No.: OF000065821 Date: 30/09/2022															
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY										PROJECT No.:															
										JOB No.:															
LOT	SPECIMEN					MINIMUM REQUIRED VALUES					OBTAINED VALUES														
	NO.	IZOD	KV	ISOV	KCU	T °C	M.Unit	MIN.	AVE.	L.E.	ABSORBED ENERGY				LATERAL EXP.				SHEAR %						
											1	2	3	AVE.	1	2	3	60	70	80	90				
L001			X			-73	J	21	27		39	40	43	41											
L001			X			-73	J	21	27		41	41	38	40											
L002			X			-73	J	21	27		37	36	35	36											
L004			X			-73	J	21	27		35	32	39	35											
L005			X			-73	J	21	27		40	43	38	40											
L006			X			-73	J	21	27		42	44	39	42											
L006			X			-73	J	21	27		43	41	41	42											
L007			X			-73	J	21	27		38	43	42	41											
L008			X			-73	J	21	27		45	39	43	42											
L009			X			-73	J	21	27		38	37	40	38											
L010			X			-73	J	21	27		37	35	41	38											
L011			X			-73	J	21	27		42	42	46	43											
L012			X			-73	J	21	27		40	39	38	39											
L012			X			-73	J	21	27		43	45	43	44											
L013			X			-73	J	21	27		39	41	44	41											
OME Metallurgica Erbesse S.r.l.										Statement															
Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it										ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION															
										Prepared  P. ROSSI															
										Approved  M. MERONI															
										F 8.2 - CMTR - 06															

INSPECTION CERTIFICATE										CERT. TYPE				CMTR Nr.				SHEET																					
HARDNESS TEST										ISO 10474 EN 10204				3.1				2203115/3 DATE 20/12/2022				5/6																	
CUSTOMER : VALVITALIA S.P.A.										ORDER No.: OF000065821 Date: 30/09/2022																													
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY										PROJECT No.:																													
										JOB No.:																													
LOT	NO. OF TEST	REQUIRED VALUES										OBTAINED VALUES																											
		AFTER HEAT TREAT. x 24h				ROOM TEMPERATURE						AFTER HEAT TREAT. x 24h				ROOM TEMPERATURE																							
		HBW	HRBW	T °C	HARD. MIN	HBW	HRBW	HRC	HV	HARDNESS		CONTR.	HARDNESS		CONTR.	HARDNESS		PL	CPL																				
										MIN	MAX		MIN	MAX		MIN	MAX		C	NC	C	NC																	
224215																																							
224215	*																																						
224230																																							
224230	*																																						
224233																																							
224233	*																																						
L001			X	590	84			X				2	91	2	18	20																							
L001	*							X						100%	210	219																							
L004								X						1	18	19																							
L004	*							X						100%	213	220																							
L005								X						1	17	18																							
L005	*							X						100%	218	224																							
L006								X						100%	222	230																							
L006			X	590	84			X				2	90	2	19	20	X																						
L007			X	590	84	X						1	90	100%	212	223	X																						
L007								X						1	18	20																							
L008			X	590	84	X						1	91	1	18	19	X																						
L008	*							X						100%	216	222																							
L009								X						100%	218	228																							
L009	*							X						1	18	20																							
L010								X						100%	216	227																							
L010	*							X						1%	19	21																							
L011			X	590	84			X				1	90	1	18	19	X																						
L011	*							X						100%	212	224																							
L012								X						2	17	18																							
L012	*							X						100%	206	215																							
L013								X						1	19	20																							
L013	*							X						100%	218	226																							
										03/02/2023																													
PL = Proof Load										CPL = Cone Proof Load										C = Conforming										NC = Not Conforming									
OME Metallurgica Erbesse S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it										Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION										Prepared  F. ROSSINI										Approved  F. VERONESE F 8.2 - CMTR - 07									

INSPECTION CERTIFICATE		CERT. TYPE	CMTR Nr.	SHEET	
Declarations		ISO 10474 EN 10204	2203115/3 DATE 20/12/2022	6/6	
CUSTOMER :VALVITALIA S.P.A.		ORDER No.OF000065821 Date: 30/09/2022:			
DESTINATION :		PROJECT : No.:			
VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY		JOB No.:			
Compliance		Material Specification			
ASTM A194-22 Ed. API SPEC 20E 2nd Ed + A1:2018 + A2:2019 + E1:2021 MACROETCH ASTM E381; (1): S1; (2): R1; (3): C2; CONFORMING DECARBURIZATION TEST ASTM F2328 / F2328M Class 3: CONFORMING - Absence of gross and partial decarburization * HARDNESS TEST PERFORMED ON 100% OF PIECES ASTM A320-22 Ed. EU DIR. 2014/68/EU (PED) REQUIREMENTS : CONFORMING ELONGATION AFTER RUPTURE IS 14% MIN. MEASURED ON Lo=5,65vSo ANSI/NACE MR-0175 / ISO 15156-2015 : CONFORMING VISUAL AND DIMENSIONAL CONTROL ON 100% OF PIECES PERFORMED WITH SATISFACTORY RESULTS. MARKING CHECK ON 100% OF PIECES: CONFORMING		VALVITALIA Spec. # MDS_013 Rev. 4 VALVITALIA Spec. # MDS_012 Rev. 4 Annex A Rev.0			
Coating		Condition			
		QTM = QUENCHED AND TEMPERED @ 1150 °F [620 °C] min Batch furnace - Furnace Calibration Acc.to API 6A 21st Ed.			
Generic Specification					
API 20E BSL-1 7M (MPS 006) - A194 API 20E-1 : Identify thread forming process: Machining/cutting L7M (MPS 006) - A320 API 20E-1 : Identify thread forming process: Rolling					
OME Metallurgica Erbesse S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it		Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION NO WELD REPAIR		Prepared  Approved  F 8.2 - CMTR - 30	



36011 Arsiero (VI) ITALY - Via Cartiera di Mezzo, 38
Tel 0445 714400 - Fax 0445 714444

Inspection Certificate

Quality Control Department

N° 13712 / 22 REV.0 Date 19/04/2023

COMPANY WITH
QUALITY SYSTEM
CERTIFIED BY DNV
= ISO 9001 & EN 9100 =
COGOLLO PLANT

36010 Cogollo del Cengio (VI) ITALY - Via M. Schiro, 2

Purchaser:

VALVITALIA S.p.A.
PIAZZA SIGMUND FREUD 1
20154 MILANO MI

Order No.:

OF000068200 REV.1

Certificate type: EN 10204 ed. MARCH 2005

☐ 2.2. ☒ 3.1. ☐ 3.2

SFRG job No.:

OV224403

Melting ☒ E.F.V.D ☐ E.F ☐ AOD ☐ VOD
practice ☐ EB ☐ VAR ☐ VIM ☐ ESR

CHEMICAL ANALYSIS / LADLE ANALYSIS

MATERIAL: ASTM-A350-LF2-CL.1 ED.18

HEAT: 289837

SUPPLIER: ACCIAIERIE VENETE S.p.A.

C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	Sn	V	Nb	N	As	H	B	Ca		
%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	ppm	%		
0,159	1,103	0,219	0,107	0,119	0,006	0,011	0,025	0,107	0,021	0,002	0,006	0,016	0,003	0,0071	0,004	1,6	1	0,0006		

C.E. (Long) = 0,388

HEAT TREATMENT

TYPE	TEMP °C	HOURS	COOLING	TYPE	TEMP °C	HOURS	COOLING
NORMALIZING	910	5	AIR				

MECHANICAL TESTS

FROM PIECE ☒

FROM QTC ☒

REQUIRED	UTS	YS 0,2	YS	EL	RA	IMPACT TEST					HB on pieces	
	MPa	MPa	MPa	%	%	Dir.	Type	Temp °C	Avr/Min	J	Min	Max
MIN	485	275		22	30	TAN	KV	-46	27	21		197
MAX	655											

Treated with the Pcs

OBTAINED

N° TEST	DIM	SECT.	LENGHT	TEMP	DIR	UTS	YS 0,2	YS	EL	RA	IMPACT TEST					HB on pieces	
	mm	mm	mm	°C		MPa	MPa	MPa	%	%	Dir.	Type	Temp °C	JOULE	Avr	Min	Max
8242	12,50	122,72	50,00	+20	TAN	504	321		32,0	73,0	TAN	KV	-46	108 95 93 99		156	163

OUR ITEM	CUSTOMER ITEM	DESCRIPTION	DRAWING	QUANTITY
34	from 340 to 340	RAW BODY 496x399x297 F	7UCP109U10..GRev0	V000043880 10
	from to			
	from to			
	from to			
	from to			
	from to			

REMARKS:

PIECES CONFORMING TO DIMENSIONAL AND VISUAL CONTROL.
MATERIAL ACCORDING TO SPEC.: MDS_001 REV. 3; QTR STOCK 58-5Q REV. 8; QTR STOCK-7U REV. 8.
MATERIAL ACCORDING TO NACE MR 01-75/ISO 15156 ed. 2015 AND MR 01.03/ISO 17945 ed. 2015.
QUALITY ASSURANCE SYSTEM IN ACCORDANCE WITH THE P.E.D. 2014/68/EU ANNEX 1 PARA. 4.3.
HEAT TREATMENT FURNACE CALIBRATED ACCORDING TO API 6D 25th ED. ANNEX H.
REDUCTION RATIO > 3:1. REF. S.N.: FROM 1 TO 10.



30 MAG. 2023

Note: Prodotti conformi ai requisiti dell'ordine - Products are in compliance with the requirements of the order - Les produits sont conforme à votre ordre - Produkt entsprechend bestellungs spezifikation.

Material conforming ☒

Material non conforming ☐

Quality Control Dept.

SIDERFORGEROSS
Q.C. Martin Diego

Surveyor:

SGQ 138/99 Page 1 di 1



 SIDERFORGEROSSİ 36011 Arsiero (VI) ITALY - Via Cartiera di Mezzo, 38 Tel.0445 714400 - Fax 0445 714444	Inspection Certificate Quality Control Department		COMPANY WITH QUALITY SYSTEM CERTIFIED BY DNV = ISO 9001 & EN 9100 = ARSIERO PLANT 36011 Arsiero (VI) ITALY - Via Cartiera di Mezzo, 38
	N° 13650 / 22 REV.0 Date 11/04/2023		

Purchaser: VALVITALIA S.P.A. PIAZZA SIGMUND FREUD 1 20154 MILANO MI	Order No.: OF000068200 REV.1	Certificate type: EN 10204 ed. MARCH 2005 ☐ 2.2. ☒ 3.1. ☐ 3.2
	SFRG job No.: OV224403	Melting practice: ☒ E.F.V.D ☐ E.F ☐ AOD ☐ VOD ☐ EB ☐ VAR ☐ VIM ☐ ESR

CHEMICAL ANALYSIS / LADLE ANALYSIS

MATERIAL: ASTM-A350-LF2-CL.1 ED.18

HEAT: 135283

SUPPLIER: Pallanzeno s.r.l.

C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	Sn	V	Nb	N	Pb	H	B	Ca		
%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	ppm	%		
0,16	1,15	0,24	0,13	0,1	0,006	0,012	0,033	0,22	0,03	0,006	0,009	0,002	0,001	0,008	0,001	1	4	0,0017		

C.E. (Long) = 0,400

HEAT TREATMENT

TYPE	TEMP °C	HOURS	COOLING	TYPE	TEMP °C	HOURS	COOLING
NORMALIZING	910	5,5	AIR				

MECHANICAL TESTS

		MPa	MPa	MPa	%	%	Dir.	Type	Temp °C	Avr/Min	J	Min	Max	
FROM PIECE	☒	MIN	485	275		22	30	TAN	KV	-46	27	21		197
FROM QTC	☒	MAX	655											

Treated with the Pcs

OBTAINED

N° TEST	DIM	SECT.	LENGHT	TEMP	DIR	UTS	YS 0,2	YS	EL	RA	IMPACT TEST							HB on pieces	
	mm	mm	mm	°C		MPa	MPa	MPa	%	%	Dir.	Type	Temp	°C	JOULE	Avr	Min	Max	
8239	12,50	122,72	50,00	+20	TAN	489	307		38,5	73,0	TAN	KV	-46		84 107 103	98	149	151	

OUR ITEM	CUSTOMER ITEM	DESCRIPTION	DRAWING	QUANTITY
30	from 300 to 300	RAW CLOSURE	V000043783	20
	from to	546x254x291 F		
	from to			
	from to			
	from to			
	from to			
	from to			

REMARKS:

PIECES CONFORMING TO DIMENSIONAL AND VISUAL CONTROL.
 MATERIAL ACCORDING TO MDS_001 Rev. 3, QTR STOCK 58-5Q REV.8 AND QTR STOCK-7U REV.8.
 MATERIAL ACCORDING TO NACE MR 01-75/ISO 15156 ed. 2015.
 QUALITY ASSURANCE SYSTEM IN ACCORDANCE WITH THE P.E.D. 2014/68/EU ANNEX 1 PARA. 4.3.
 HEAT TREATMENT FURNACE CALIBRATED ACCORDING TO API 6D 24TH ED. ANNEX F SUITABLE TO ACHIEVE THE REQUIREMENTS OF API 6D 25TH ED. ANNEX H.
 REDUCTION RATIO >3:1. REF. S.N. 1 TO 20.



04 MAG. 2023

Note: Prodotti conformi ai requisiti dell'ordine - Products are in compliance with the requirements of the order - Les produits sont conforme à votre ordre - Produkt entsprechend bestellungs spezifikation.

Material conforming ☒ Quality Control Dept. *Diego Martini*
 Material non conforming ☐ Surveyor:

SGQ 138/99 Page 1 di 1





36011 Arsiero (VI) ITALY - Via Cartiera di Mezzo, 38
Tel. 0445 714400 - Fax 0445 714444

Inspection Certificate

Quality Control Department

N° 13695 / 22 REV.0 Date 18/04/2023

COMPANY WITH
QUALITY SYSTEM
CERTIFIED BY DNV
= ISO 9001 & EN 9100 =
BUSANO PLANT

10080 Busano Canavese (TO) ITALY - Via Rolando Perino, 37

Purchaser:

VALVITALIA S.P.A.
PIAZZA SIGMUND FREUD 1
20154 MILANO MI

Order No.:

OF000068200 REV.1

Certificate type: EN 10204 ed. MARCH 2005

☐ 2.2. ☐ 3.1. ☐ 3.2

SRFG job No.:

OV224403

Melting

practice ☐ E.F.V.D

☒ E.F

☐ AOD

☐ VOD

☐ EB

☐ VAR

☐ VIM

☐ ESR

CHEMICAL ANALYSIS / LADLE ANALYSIS

MATERIAL: ASTM-A350-LF2-CL.1

HEAT: 288886

SUPPLIER: ACCIAIERIE VENETE S.p.A.

C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	Sn	V	Nb	N	Pb	As	H	B	Ca	
%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%
0,154	1,133	0,226	0,126	0,113	0,005	0,015	0,03	0,186	0,032	0,009	0,01	0,006	0,001	0,0056	0,0007	0,005	1,6	0,0002	0,0008	

C.E. (Long) = 0,394

HEAT TREATMENT

TYPE	TEMP °C	HOURS	COOLING	TYPE	TEMP °C	HOURS	COOLING
NORMALIZING	910	5	AIR				

MECHANICAL TESTS

REQUIRED

UTS YS 0,2 YS

EL RA

IMPACT TEST

HB on pieces

FROM PIECE ☐

FROM QTC ☒

MIN

485 275

22 30

LNG

KV

-46

27

21

Min

Max

MAX

580

Treated with the Pcs

OBTAINED

N° TEST	DIM	SECT.	LENGHT	TEMP	DIR	UTS	YS 0,2	YS	EL	RA	IMPACT TEST								HB on pieces	
	mm	mm	mm	°C		MPa	MPa	MPa	%	%	Dir.	Type	Temp °C	JOULE	Avr	Min	Max			
8537	12,50	122,72	50,00	+20	LNG	503	327		36,0	75,0	LNG	KV	-46	53 59 66 59		148	152			

OUR CUSTOMER
ITEM ITEM

DESCRIPTION

DRAWING

QUANTITY

24	from 240	to 240	SFERA VALVOLA SFERA TYPE 5U GREZ. 10"	5USF101U00..GRev0	V000042485	10
	from	to	387x254x288 F			
	from	to				
	from	to				
	from	to				
	from	to				
	from	to				

REMARKS:

PIECES CONFORMING TO DIMENSIONAL AND VISUAL CONTROL.

MATERIAL ACCORDING TO SPEC.: MDS_001 REV. 3; QTR STOCK 58-5Q REV. 8; QTR STOCK-7U REV. 8.

MATERIAL ACCORDING TO NACE MR 01-75/ISO 15156 ed. 2015 AND MR 01.03/ISO 17945 ed. 2015.

QUALITY ASSURANCE SYSTEM IN ACCORDANCE WITH THE P.E.D. 2014/68/EU ANNEX 1 PARA. 4.3.

FURNACE CALIBRATED ACCORDING TO ANNEX M API 6A 21st Ed. AND ANNEX H API 6D 25th Ed.

REDUCTION RATIO > 3:1. FOR CHEMICAL ANALYSIS SEE FOGERROSSI RESEARCH REPORT No L230567/A.



04 MAG. 2023

Note: Prodotti conformi ai requisiti dell'ordine - Products are in compliance with the requirements of the order - Les produits sont conforme à votre ordre - Produkt entsprechend bestellungs spezifikation.

Material conforming ☒

Quality Control Dept.:

Surveyor:

Material non conforming ☐

SIDERFORGEROSS
C. FENOGLIO EMILIO

SGQ 138/99 Page 1 di 1



Stem: R2226600040001, R2226600040002

DC0202 – Certificato di prodotto Rev. 2



CERTIFICATO : OR23 – 3881
CERTIFICATE
EN 10204 3.1



Cliente : VALVITALIA S.p.A. Customer	Consegna : DDT 230367 of 31.05.2023 Delivery
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Vs Ordine : OF000068383-1 of 11.01.2023 Your order Pos. 130 - COD. V000146704 - COMM. R222667U10	Ns Ordine : 230124 of 17.03.2023 Our order
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Descrizione materiale : ROUND BAR PEELED AISI 4140 / 42CrMo4 HARDENED Material description NACE MR0175/ISO 15156 UNS G41400

Grado : 4140 Grade	Diametro : 110 mm. Diameter	Colata : 222248 Heat	Quantità : kg. 189,00 Quantity
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ANALISI DI COLATA / HEAT ANALYSIS

C	Si	Mn	P	S	Cr	Ni	Mo	Al	Cu	V	Nb
.435	.200	.820	.009	.023	1.180	.180	.170	.016	.240	.004	.004

PROVE MECCANICHE / MECHANICAL TESTS

Prove longitudinali eseguite +20°C alle condizioni di fornitura
 Longitudinal testing at +20°C in delivery condition

Snervamento Rp 0,2% : 539 N/mm ² Yield strength	Rottura Rm : 704 N/mm ² Tensile strength
Allungamento A5D % : 23 Elongation	Strizione Z% : 56.9 Striction
Resilienza KV -46°C : 40 - 38.5 - 32.7 J (Average 37.1 J) Impact value	Durezza : 211 HB Hardness
Controllo spettrografico : N.D. Spectrographic examination	Ultrasuoni : satisfactory Ultrasonic test
Dimensione grano : 7 Grain size	Corrosione : N.D. Corrosion Test
Trattamento termico : Quenching 860°C and Heat treatment tempering 630°C	Controllo visivo e dimensionale : satisfactory Visual and dimensional examination
Responsabile Qualita' : Quality control	Data : 31.05.2023 Date



19 GIU. 2023

Seats: R2226600040001, R2226600040002

 IRE- OMBA S.p.a ISO 9001:2015 ISO 14001:2015 ISO 45001:2018	Test certificate / Certificato Prove Materiale 3.1 EN 10204:2004	
	IRE1221156/5 	N° - Rev 1221156/5
Customer/Ciente: BBD S.R.L.		

Order/Ordine: **2022 AF 000645**

Inspection/Collaudo: IRE OMBA

Job/Comm.: **1221156**

Mater./Spec.:

Forging ratio / Rapporto di Fucinatura >= 4:1

Manufactur. process: forged+rolled rings

Manufactur. process: Forged + rolled rings

LF2 CL1 ASTM A350/18

Prova n° Test-Batch	Item Pos.	Pieces n° Pezzi	ø est mm	ø int mm	h mm	Identification/Identificazione	HBW1	HBW2
153468	010	10	320	275	68	G-AMA500-0009 2022 AF 000645	145	
153468	080	37	339	305	56	G-ANA500-1274 2022 AF 000645	148	145
153468	100	10	320	275	68	G-AMA500-0009 2022 AF 000645	145	
153468	110	8	320	275	68	G-AMA500-0009 2022 AF 000645	145	
153468	300	20	289	254	55	G-ANA500-1120 2022 AF 000645	145	148
153468	310	20	289	254	55	G-ANA500-1120 2022 AF 000645	145	148
153468	340	40	344	305	56	G-ANA500-1153 2022 AF 000645	145	148
153468	400	30	339	305	56	G-ANA500-1274 2022 AF 000645	145	148

Chemical analysis % / Analisi chimica

Heat n° / Colata

Steel mill / Acciaieria

292825**ACC. VENETE S.P.A. - ITALY-(IT)**Cert. orig. **615465**

Vacuum degassed

C	Mn	Si	P	S	Cu	Cr	Ni	Mo	Al
0.1630	1.1570	0.2120	0.0100	0.0070	0.1100	0.0930	0.1120	0.0300	0.0220
V	Nb	Ti	Sn	B	H	N	Ceq	P+S	Nb+V+Ti
0.0050	0.0010	0.0080	0.0060	0.0003	0.00016	0.0058	0.396	0.017	0.014
Cu+6Sn	Cr+Mo	CuCrMo	CuNiCrMo	CuNiCrMoV	NiCrMo	Nb+V			
0.146	0.123	0.233	0.345	0.35	0.235	0.006			

Heat treatment / Trattamento termico

Normalized 900°C 1.5h Air

Furnace N°:TTS_F1

Due date: 15/03/2023

Furnace Calibration Doc. n°:22-0106\006

Tension Test/Trazione +20°C ASTM A370/ ASTM E8 ø=12.5mm Lo=50mm

		Required values/Valori richiesti											
		Rp .2% N/mm2		Rp 1% N/mm2		Rm N/mm2		A4d %		A5d %		Z %	
Sense/Senso		min	max	min	max	min	max	min	max	min	max	min	max
TANG		275				485	655					30	
		Obtained values/Valori rilevati											
Test n°	Sense/Senso	Rp .2% N/mm2		Rp 1% N/mm2		Rm N/mm2		A4d %		A5d %		Z %	
153468	TANG	312				516		32.5				67.4	

Impact test/Resilienza ASTM A370/ ASTM E23 10x10mm

Required values/Valori richiesti						Obtained values/Valori rilevati				
Sense/Senso	Type/tipo	T°C	Med./Av.	Min.	U.M	Test/Prova	Val1	Val2	Val3	Med./Av.
TANG	KV	-46	27	21	J	153468	85	90	82	85.67

Hardness test / Durezza: ASTM E10

Required values/Valori richiesti				Obtained values/Valori rilevati		
Min	Max	Type/tipo		Test/Prova	Val1	Val2
	197	HBW		153468	145	150

BBD S.r.l.
 Controllo Qualità
 F. B. 13/04/2023



21 APR. 2023

Data / Date (d/m/y) 12/01/23		Q.C DEPT. <i>Laura Gamba</i> GAMBA LAURA	Pag. 1 di 2
IRE - OMBA Spa Via Carlo Serassi, 14 24100 BERGAMO ITALY BG Te.: 035 / 41 60 811 Fax.: 035 / 41 60 856			

 IRE- OMBA S.p.a ISO 9001:2015 ISO 14001:2015 ISO 45001:2018	Test certificate / Certificato Prove Materiale 3.1 EN 10204:2004	
	IRE12211156/5 	N° - Rev 12211156/5

Customer/Cliente: **BBD S.R.L.**

Order/Ordine: **2022 AF 000645**

Inspection/Collaudo: IRE OMBA

Job/Comm.: **12211156**

Spec./Note

MDS_001 Rev.3

API 6D 25th Edition Annex H

NACE MR0175/ISO15156-2 Ed.15

IR22099-01 R.4

Note



21 APR. 2023

Data / Date (d/m/y) 12/01/23	Q.C DEPT. <i>Laura Gamba</i>  GAMBA LAURA	Pag. 2 di 2
IRE - OMBA Spa Via Carlo Serassi, 14 24100 BERGAMO ITALY BG Te.: 035 / 41 60 811 Fax.: 035 / 41 60 856		

 IRE- OMBA S.p.a ISO 9001:2015 ISO 14001:2015 ISO 45001:2018	Test certificate / Certificato Prove Materiale 3.1 EN 10204:2004	
	IRE12211156/18 	N° - Rev 12211156/18

Customer/Ciente: **BBD S.R.L.**

Order/Ordine: **2022 AF 000645**

Inspection/Collaudo: IRE OMBA

Job/Comm.: **12211156**

Mater./Spec.:

Forging ratio / Rapporto di Fucinatura >= 4:1

LF2 CL1 ASTM A350/18

Manufactur. process: forged+rolled rings

Manufactur. process: Forged + rolled rings

Prova n° Test-Batch	Item Pos.	Pieces n° Pezzi	ø est mm	ø int mm	h mm	Identification/Identificazione	HBW1	HBW2
153741	110	1	320	275	68	G-AMA500-0009 2022 AF 000645	145	

Chemical analysis % / Analisi chimica

Heat n° / Colata			Steel mill / Acciaieria						
292825			ACC. VENETE S.P.A. - ITALY-(IT)						
			Cert. orig. 615465						
			Vacuum degassed						
C	Mn	Si	P	S	Cu	Cr	Ni	Mo	Al
0.1630	1.1570	0.2120	0.0100	0.0070	0.1100	0.0930	0.1120	0.0300	0.0220
V	Nb	Ti	Sn	B	H	N	Ceq	P+S	Nb+V+Ti
0.0050	0.0010	0.0080	0.0060	0.0003	0.00016	0.0058	0.396	0.017	0.014
Cu+6Sn	Cr+Mo	CuCrMo	CuNiCrMo	CuNiCrMoV	NiCrMo	Nb+V			
0.146	0.123	0.233	0.345	0.35	0.235	0.006			

Heat treatment / Trattamento termico

Normalized 900°C 1.5H Air

Tension Test/Trazione +20°C ASTM A370/ ASTM E8 ø=12.5mm Lo=50mm													
		Required values/Valori richiesti											
		Rp .2% N/mm2		Rp 1% N/mm2		Rm N/mm2		A4d %		A5d %		Z %	
		Sense/Senso	min	max	min	max	min	max	min	max	min	max	min
TANG		275				485	655	22				30	
		Obtained values/Valori rilevati											
Test n°	Sense/Senso	Rp .2% N/mm2		Rp 1% N/mm2		Rm N/mm2		A4d %		A5d %		Z %	
153741	TANG	307				511		32.6				67.5	

Impact test/Resilienza ASTM A370/ ASTM E23 10x10mm										
Required values/Valori richiesti						Obtained values/Valori rilevati				
Sense/Senso	Type/tipo	T°C	Med./Av.	Min.	U.M	Test/Prova	Val1	Val2	Val3	Med./Av.
TANG	KV	-46	27	21	J	153741	88	99	91	92.67

Hardness test / Durezza: ASTM E10										
Required values/Valori richiesti						Obtained values/Valori rilevati				
Min	Max	Type/tipo				Test/Prova	Val1	Val2		
	197	HBW				153741	143	148		

Spec./Note

MDS_001 Rev.3

API 6D 25th Edition Annex H

NACE MR0175/ISO15156-2 Ed.15

IR22099-01 R.4

Note



21 APR. 2023

BBD S.r.l.
Controllo Qualità
Firma: *[Signature]* 20/04/2023

Data / Date (d/m/y) 25/01/23		Q.C DEPT. <i>Laura Gamba</i> GAMBA LAURA	Pag. 1 di 1
IRE - OMBA Spa Via Carlo Serassi, 14 24100 BERGAMO ITALY BG Te.: 035 / 41 60 811 Fax.: 035 / 41 60 856			

 IRE- OMBA S.p.a ISO 9001:2015 ISO 14001:2015 ISO 45001:2018	Test certificate / Certificato Prove Materiale 3.1 EN 10204:2004	
	IRE12211156/20 	N° - Rev 12211156/20
Customer/Cliente: BBD S.R.L.		

Order/Ordine: **2022 AF 000645**

Inspection/Collaudo: IRE OMBA

Job/Comm.: **12211156**

Mater./Spec.:

Forging ratio / Rapporto di Fucinatura >= 4:1

LF2 CL1 ASTM A350/18

Manufactur. process: forged+rolled rings

Manufactur. process: Forged + rolled rings

Prova n° Test-Batch	Item Pos.	Pieces n° Pezzi	ø est mm	ø int mm	h mm	Identification/Identificazione	HBW1	HBW2
154197	110/1	1	320	275	68	G-AMA500-0009 2022 AF 000645	143	

Chemical analysis % / Analisi chimica

Heat n° / Colata

Steel mill / Acciaieria

290249

ACC. VENETE S.P.A. - ITALY-(IT)

Cert. orig. **591918**

Vacuum degassed

C	Mn	Si	P	S	Cu	Cr	Ni	Mo	Al
0.16	1.163	0.24	0.011	0.0070	0.157	0.12	0.133	0.036	0.024
V	Nb	Ti	Sn	B	H	N	Ceq	P+S	Nb+V+Ti
0.0040	0.0010	0.0080	0.0070	0.0004	0.00015	0.0070	0.405	0.018	0.013
Cu+6Sn	Cr+Mo	CuCrMo	CuNiCrMo	CuNiCrMoV	NCrMo	Nb+V			
0.199	0.156	0.313	0.446	0.45	0.289	0.005			

Heat treatment / Trattamento termico

Normalized 900°C 1.5H Air

Tension Test/Trazione +20°C ASTM A370/ ASTM E8 ø=12.5mm Lo=50mm														
		Required values/Valori richiesti												
		Rp .2% N/mm2		Rp 1% N/mm2		Rm N/mm2		A4d %		A5d %		Z %		
		Sense/Senso	min	max	min	max	min	max	min	max	min	max	min	max
		TANG	275				485	655	22				30	
		Obtained values/Valori rilevati												
Test n°	Sense/Senso	Rp .2% N/mm2		Rp 1% N/mm2		Rm N/mm2		A4d %		A5d %		Z %		
154197	TANG	322				504		29.9				73.5		

Impact test/Resilienza ASTM A370/ ASTM E23 10x10mm									
Required values/Valori richiesti						Obtained values/Valori rilevati			
Sense/Senso	Type/tipo	T°C	Med./Av.	Min.	U.M	Test/Prova	Val1	Val2	Med./Av.
TANG	KV	-46	27	21	J	154197	78	72	81

Hardness test / Durezza: ASTM E10									
Required values/Valori richiesti						Obtained values/Valori rilevati			
Min	Max	Type/tipo				Test/Prova	Val1	Val2	
	197	HBW				154197	143	146	

Spec./Note									
MDS_001 Rev.3					API 6D 25th Edition Annex H				
NACE MR0175/ISO15156-2 Ed.15					IR22099-01 R.4				

Note

BBD S.r.l.
 Controllo Qualità
 Firmato C. G. 13/02/2023



21 APR. 2023

Data / Date (d/m/y) 10/02/23	Q.C DEPT. <i>Laura Gamba</i> GAMBA LAURA	Pag. 1 di 1
IRE - OMBA Spa Via Carlo Serassi, 14 24100 BERGAMO ITALY BG Te.: 035 / 41 60 811 Fax.: 035 / 41 60 856		



CERTIFICATO : OR23 - 5826/3
CERTIFICATE
EN 10204 3.1



Cliente : VALVITALIA S.p.A. Customer	Consegna : DDT 230393 of 08.06.2023 Delivery
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Vs Ordine : OF000068383-1 of 11.01.2023 Your order Pos. 100 - COD. V000138473 - COMM. R222667U10	Ns Ordine : 230124 of 17.03.2023 Our order
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Descrizione materiale : ROUND BAR FORGED LF2 CL.1 Material description ASTM A350 - ISO 15156/NACE MR0175
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Grado : LF2 Grade	Diametro : 170 mm. Diameter	Colata : 202543 Heat	Quantità : kg. 111,00 Quantity
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ANALISI DI COLATA / HEAT ANALYSIS

C	Si	Mn	P	S	Cr	Ni	Mo	Al	Cu	V	Nb	Sn	H (ppm)	Ceq
.170	.250	1.110	.007	.003	.110	.150	.040	.034	.170	.005	.002	.010	1.200	.407

PROVE MECCANICHE / MECHANICAL TESTS

Prove longitudinali eseguite +20°C alle condizioni di fornitura
 Longitudinal testing at +20°C in delivery condition

Snervamento Rp 0,2% : 359 N/mm2 Yield strength	Rottura Rm : 540 N/mm2 Tensile strength
Allungamento A % : 32 Elongation	Strizione Z% : 51 Striction
Resilienza KV -46 °C : 81 - 97 - 93 J (average 90.3 J) Impact value	Durezze : 158 - 162 HB Hardness
Controllo spettrografico : N.D. Spectrographic examination	Ultrasuoni : N.D. Ultrasonic test
Dimensione grano : 7 ASTM E112 Grain size	Rapporto di riduzione : 10 Reduction ratio
Trattamento termico : normalized 900°C/6h/air Heat treatment	Controllo visivo e dimensionale : satisfactory Visual and dimensional examination
Responsabile Qualità' : Quality control	Data : 08.06.2023 Date



19 GIU. 2023

INSPECTION CERTIFICATE				CERT. TYPE		CMTR Nr.	SHEET			
LIST OF SUPPLIED PRODUCTS				ISO 10474		2300494/4	1/6			
				EN 10204		DATE 05/05/2023				
CUSTOMER : VALVITALIA S.P.A.				ORDER No.: OF000069504 REV.1 Date: 31/03/2023						
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY				PROJECT No.:						
				JOB No.:						
LOT	OME ITEM	P.O. ITEM	Q TY	Dimension	Drawing No.	Spec / Grade	Heat No.	Marking	Cond.	Coat.
L001	200	20	160 No.	1 " 8 UNRC x 145	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	210016	OME IT-L7M 20E1-729	QTM	
L001	300	30	320 No.	1 " 8 UNRC x 135	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	210016	OME IT-L7M 20E1-730	QTM	
L002	1000	100	840 No.	1.1/8 " 8 UN	ASME B18.2.2-T.1.1.1-11 Ed 22	ASTM A 194 / A 194M API 20E-1 GR.7M	18/70440	OME IT-D 7ML 20E1-737	QTM	
L003	1100	110	240 No.	1.1/8 " 8 UNR x 165	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	212046	OME IT-L7M 20E1-738	QTM	
L003	1200	120	600 No.	1.1/8 " 8 UNR x 155	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	212046	OME IT-L7M 20E1-739	QTM	
L003	1400	140	280 No.	1.1/8 " 8 UNR x 145	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	212046	OME IT-L7M 20E1-741	QTM	
L006	1600	160	20 No.	1 " 8 UNRC x 48	ASME B18.3-2012 TAB 1	ASTM A 320 / A 320M API 20E-1 L7M	267387	OME IT-L7M 20E1-743	QTM	
L002	1700	170	360 No.	1.1/8 " 8 UN	ASME B18.2.2-T.1.1.1-11 Ed 22	ASTM A 194 / A 194M API 20E-1 GR.7M	18/70440	OME IT-744	QTM	
L007	1800	180	480 No.	1.3/8 " 8 UNR x 152	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	211865	OME IT-L7M 20E1-745	QTM	
L008	2200	220	640 No.	1.1/4 " 8 UN	ASME B18.2.2-T.1.1.1-11 Ed 22	ASTM A 194 / A 194M API 20E-1 GR.7M	16/78540	OME IT-7LM/7ML 20E1-749 A10	QTM	
L009	2300	230	480 No.	1.1/4 " 8 UNR x 170	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	218952	OME IT-L7M 20E1-750	QTM	
L010	2800	280	720 No.	1.3/8 " 8 UN	ASME B18.2.2-T.1.1.1-11 Ed 22	ASTM A 194 / A 194M API 20E-1 GR.7M	1B0002309	OME IT-D 7ML 20E1 755 132	QTM	
L007	2900	290	240 No.	1.3/8 " 8 UNR x 168	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	211865	O-756	QTM	
L012	3300	330	160 No.	2 1/2 " 8 UNR x 255	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	218714	OME IT-L7M 20E1-760	QTM	
L013	3400	340	120 No.	1.3/4 " 8 UNR x 180	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	BD4395	OME IT-L7M 20E1-761	QTM	
L013	3900	390	160 No.	1.3/4 " 8 UNR x 185	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	BD4395	OME IT-L7M 20E1-766	QTM	
L001	4100	410	100 No.	1 " 8 UNRC x 140	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	210016	OME IT-L7M 20E1-768	QTM	
L015	4200	420	80 No.	2 " 8 UNR x 220	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	217995	OME IT-L7M 20E1-769	QTM	
L015	4300	430	80 No.	2 " 8 UNR x 210	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	217995	OME IT-L7M 20E1-770	QTM	
L017	4500	450	140 No.	1.1/2 " 8 UN	ASME B18.2.2-T.1.1.1-11 Ed 22	ASTM A 194 / A 194M API 20E-1 GR.7M	1B0004936	OME IT-D 271 7ML 20E1 772	QTM	
L009	4700	470	160 No.	1.1/4 " 8 UNR x 185	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	218952	OME IT-L7M 20E1-774	QTM	
L001	4800	480	300 No.	1 " 8 UNRC x 133	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	210016	OME IT-L7M 20E1-775	QTM	
										
VISUAL / DIMENSIONAL EXAMINATION : Conforming										
OME Metallurgia Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it				Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION				Prepared 		Approved 
F 8.2 - CMTR - 20										

INSPECTION CERTIFICATE				CERT. TYPE				CMTR Nr.				SHEET							
HEAT ANALYSIS				ISO 10474 EN 10204				3.1 DATE 05/05/2023				2/6							
CUSTOMER : VALVITALIA S.P.A.				ORDER No.: OF000069504 REV.1 Date: 31/03/2023															
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY				PROJECT No.:															
				JOB No.:															
LOT	Heat No.	C	Mn	Si	P	S	Cr	Ni	Mo	V	Ti	Cu	W	Al	B	Nb	Co		
		Zn	Pb	Sn	Be	O	N	H	Fe	Nb+Ta	Cb	Al+Ti	Cb+Ta	Se	Bi	Ca	Mg		
L001	210016	0,4100	0,7500	0,2700	0,0100	0,0010	0,9600		0,2000										
L002	18/70440	0,3800	0,8200	0,2500	0,0210	0,0150	1,0000		0,1900										
L003	212046	0,3900	0,8700	0,3300	0,0100	0,0010	1,0000		0,2210										
L006	267387	0,3940	0,8500	0,2800	0,0090	0,0030	1,0100		0,2300										
L007	211865	0,4100	0,8800	0,3400	0,0120	0,0020	1,0300		0,2100										
L008	16/78540	0,3950	0,8300	0,1800	0,0250	0,0220	0,9100		0,1900										
L009	218952	0,4100	0,8800	0,3300	0,0120	0,0010	1,0200		0,2140										
L010	1B0002309	0,4000	0,8420	0,1890	0,0100	0,0070	0,9730		0,2170										
L012	218714	0,4100	0,8800	0,3400	0,0100	0,0020	1,0200		0,2310										
L013	BD4395	0,4100	0,8500	0,1900	0,0130	0,0050	1,0200		0,1700										
L015	217995	0,4000	0,8700	0,3500	0,0110	0,0010	1,0300		0,2050										
L017	1B0004936	0,4070	0,8740	0,2140	0,0110	0,0080	1,0180		0,1840										
 17 MAG. 2023																			
OME Metallurgica Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it							Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION							Prepared 		 F 8.2 - CMTR - 29			

INSPECTION CERTIFICATE						CERT. TYPE		CMTR Nr.		SHEET					
TENSILE TEST						ISO 10474 EN 10204 3.1		2300494/4 DATE 05/05/2023		3/6					
CUSTOMER : VALVITALIA S.P.A.						ORDER No.: OF000069504 REV.1 Date: 31/03/2023									
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY						PROJECT No.: JOB No.:									
LOT	SPECIMEN				T °C	Measure Unit	REQUIRED VALUES					OBTAINED VALUES			
	NO.	DIA. mm	AREA mm2	L mm			MIN Rp 0.2	R MIN MAX		MIN E %	MIN R.A. %	Rp 0.2	R	E %	R.A. %
L001		12,5	122,66	50,00	Room	MPa	550	690		18	50	612	725	23,1	64,0
L003		12,5	122,66	50,00	Room	MPa	550	690		18	50	620	729	25,2	63,5
L006		12,5	122,66	50,00	Room	MPa	550	690		18	50	608	730	22,3	65,0
L007		12,5	122,66	50,00	Room	MPa	550	690		18	50	611	722	24,1	66,1
L009		12,5	122,66	50,00	Room	MPa	550	690		18	50	626	713	23,7	63,0
L012		12,5	122,66	50,00	Room	MPa	550	690		18	50	606	711	23,4	65,7
L013		12,5	122,66	50,00	Room	MPa	550	690		18	50	622	730	25,0	63,1
L015		12,5	122,66	50,00	Room	MPa	550	690		18	50	625	709	26,3	65,2
OME Metallurgica Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it						Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION						Prepared 			

VALVITALIA S.P.A.
17 MAR. 2023

INSPECTION CERTIFICATE										CERT. TYPE		CMTR Nr.		SHEET									
IMPACT TEST										ISO 10474 3.1 EN 10204		2300494/4		4/6									
CUSTOMER : VALVITALIA S.P.A.										ORDER No.: OF000069504 REV.1 Date: 31/03/2023													
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY										PROJECT No.:													
										JOB No.:													
LOT	SPECIMEN					MINIMUM REQUIRED VALUES					OBTAINED VALUES												
	NO.	IZOD	KV	ISOV	KCU	T °C	M.Unit	MIN.	AVE.	L.E.	ABSORBED ENERGY				LATERAL EXP.			SHEAR %					
											1	2	3	AVE.	1	2	3	60	70	80	90		
L001			X			-73	J	21	27		39	41	43	41									
L002			X			-73	J	21	27		42	44	38	41									
L002			X			-73	J	21	27		41	45	42	43									
L003			X			-73	J	21	27		40	43	39	41									
L006			X			-73	J	21	27		39	38	41	39									
L007			X			-73	J	21	27		44	42	40	42									
L008			X			-73	J	21	27		38	44	41	41									
L009			X			-73	J	21	27		43	45	39	42									
L010			X			-73	J	21	27		41	40	40	40									
L012			X			-73	J	21	27		43	37	44	41									
L013			X			-73	J	21	27		40	43	41	41									
L015			X			-73	J	21	27		44	41	38	41									
L017			X			-73	J	21	27		42	38	44	41									
OME Metallurgia Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it										Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION										Prepared  F. ROSSI		 F. ROSSI	

VALVITALIA S.P.A.
17 MAG. 2023

INSPECTION CERTIFICATE										CERT. TYPE		CMTR Nr.		SHEET																									
HARDNESS TEST										ISO 10474 3.1 EN 10204		2300494/4		5/6																									
CUSTOMER : VALVITALIA S.P.A.										ORDER No.: OF000069504 REV.1 Date: 31/03/2023																													
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY										PROJECT No.:																													
										JOB No.:																													
LOT	NO. OF TEST	REQUIRED VALUES										OBTAINED VALUES																											
		AFTER HEAT TREAT. x 24h				ROOM TEMPERATURE						AFTER HEAT TREAT. x 24h				ROOM TEMPERATURE																							
		HBW	HRBW	T °C	HARD. MIN	HBW	HRBW	HRC	HV	HARDNESS		CONTR. QTY	HARDNESS		CONTR. QTY	HARDNESS		PL		CPL																			
										MIN	MAX		MIN	MAX		MIN	MAX	C	NC	C	NC																		
L001							X		22						19	21																							
L001	*					X			200	235				100%	219	228																							
L002		X		590	84		X		22		2	91		2	19	20	X																						
L002	*					X			159	235				100%	216	226																							
L003						X			200	235				1	206	213																							
L003							X		22					1	18	19																							
L006							X		22					1	18	20																							
L006	*					X			200	235				100%	210	221																							
L007							X		22					1	17	18																							
L007	*					X			200	235				100%	205	213																							
L008		X		590	84		X		22		1	90		1	18	20	X																						
L008	*					X			159	235				100%	212	222																							
L009						X			200	235				1	216	224																							
L009							X		22					1	18	20																							
L010		X		590	84		X		22		1	90		1	18	19	X																						
L010	*					X			159	235				100%	210	223																							
L012							X		22					1	19	20																							
L012	*					X			200	235				100%	221	230																							
L013							X		22					1	17	19																							
L013	*					X			200	235				100%	209	217																							
L015							X		22					1	19	21																							
L015	*					X			200	235				100%	220	230																							
L017		X		590	84		X		22		1	90		1	17	18	X																						
L017	*					X			159	235				100%	215	222																							
										 17 MAG. 2023																													
PL = Proof Load										CPL = Cone Proof Load										C = Conforming										NC = Not Conforming									
OME Metallurgica Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it										Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION										Prepared  G. ROSSINI										 F. MERONI									
																				F 8 2 - CMTR - 07																			

INSPECTION CERTIFICATE		CERT. TYPE ISO 10474 3.1 EN 10204	CMTR Nr. 2300494/4 DATE 05/05/2023	SHEET 6/6	
Declarations					
CUSTOMER :VALVITALIA S.P.A.		ORDER No.OF000069504 REV.1 Date: 31/03/2023:			
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY		PROJECT : No.:			
		JOB No.:			
Compliance ASTM A320-22 Ed. API SPEC 20E 2nd Ed + A1:2018 + A2:2019 + E1:2021 MACROETCH ASTM E381; (1): S1; (2): R1; (3): C2; CONFORMING DECARBURIZATION TEST ASTM F2328 / F2328M Class 3: CONFORMING - Absence of gross and partial decarburization * HARDNESS TEST PERFORMED ON 100% OF PIECES ASTM A194-22 Ed. ANSI/NACE MR-0175 / ISO 15156-2015 : CONFORMING		Material Specification VALVITALIA MDS 013 Rev.4 VALVITALIA MDS 012 Rev.4 VALVITALIA Annex A Rev.0			
Coating		Condition QTM = QUENCHED AND TEMPERED @ 1150 °F [620 °C] min Batch furnace - Furnace Calibration Acc.to API 6A 21st Ed.			
Generic Specification API 20E BSL-1 L7M (MPS 006) - A320 API 20E-1 : Identify thread forming process: Rolling 7M (MPS 006) - A194 API 20E-1 : Identify thread forming process: Machining/cutting					
OME Metallurgia Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it		Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION NO WELD REPAIR		Prepared  17 MAG. 2023  F 8.2 - CMTR - 30	

INSPECTION CERTIFICATE				CERT. TYPE		CMTR Nr.	SHEET			
LIST OF SUPPLIED PRODUCTS				ISO 10474		2300494/10	1/6			
				EN 10204		DATE 16/05/2023				
CUSTOMER : VALVITALIA S.P.A.				ORDER No.: OF000069504 REV 1 Date: 31/03/2023						
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY				PROJECT No.:						
				JOB No.:						
LOT	OME ITEM	P.O. ITEM	Q.TY	Dimension	Drawing No.	Spec / Grade	Heat No.	Marking	Cond.	Coat.
L001	900	90	2240 No.	1.1/8 " 8 UN	ASME B18.2.2-T.1.1.1-11 Ed 22	ASTM A 194 / A 194M API 20E-1 GR. 7M	22/74995	OME IT-7ML 20E1-736	QTM	
L002	1300	130	480 No.	1 " 8 UNC	ASME B18.2.2-T.1.1.1-11 Ed 22	ASTM A 194 / A 194M API 20E-1 GR. 7M	18/70440	HEX OME IT 7ML/ 7LM A8 20E1 740	QTM	
L003	1500	150	80 No.	1.1/8 " 8 UNR x 160	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	212046	OME IT-L7M 20E1-742	QTM	
L004	2000	200	200 No.	M16 x 2 x 70	UNI 6610	ASTM A 320 / A 320M API 20E-1 L7M	225006	O IT-L7M- 747	QTM	
L002	3000	300	400 No.	1 " 8 UNC	ASME B18.2.2-T.1.1.1-11 Ed 22	ASTM A 194 / A 194M API 20E-1 GR. 7M	18/70440	OME IT 7ML\ 7LM 20E1 757 A8 HEX	QTM	
L005	3800	380	110 No.	1.3/4 " 8 UN	ASME B18.2.2-T.1.1.1-11 Ed 22	ASTM A 194 / A 194M API 20E-1 GR. 7M	16/78540	OME IT-7ML 20E1-765	QTM	
L006	3800	380	170 No.	1 3/4 " 8 UN	ASME B18.2.2-T.1.1.1-11 Ed 22	ASTM A 194 / A 194M API 20E-1 GR. 7M	1B0002308	OME IT-7ML 20E1-765	QTM	
L007	4000	400	160 No.	2 1/2 " 8 UNR x 265	ANSI B16.5	ASTM A 320 / A 320M API 20E-1 L7M	218714	OME IT-L7M 20E1-767	QTM	
 29 MAG. 2023										
VISUAL / DIMENSIONAL EXAMINATION : Conforming										
OME Metallurgica Erbesse S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it				Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION			Prepared 		Approved 	
							F 82 - CMTR - 20			

INSPECTION CERTIFICATE							CERT. TYPE		CMTR Nr.		SHEET										
HEAT ANALYSIS							3.1		2300494/10		2/6										
CUSTOMER : VALVITALIA S.P.A.							ORDER No.: OF000069504 REV.1 Date: 31/03/2023														
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY							PROJECT No.:														
							JOB No.:														
LOT	Heat No.	C	Mn	Si	P	S	Cr	Ni	Mo	V	Ti	Cu	W	Al	B	Nb	Co				
		Zn	Pb	Sn	Be	O	N	H	Fe	Nb+Ta	Cb	Al+Ti	Cb+Ta	Se	Bi	Ca	Mg				
L001	22/74995	0,4050	0,8400	0,2600	0,0220	0,0030	1,0500		0,2000												
L002	18/70440	0,3800	0,8200	0,2500	0,0210	0,0150	1,0000		0,1900												
L003	212046	0,3900	0,8700	0,3300	0,0100	0,0010	1,0000		0,2210												
L004	225006	0,4000	0,8100	0,2700	0,0120	0,0020	1,0300		0,2050												
L005	16/78540	0,3950	0,8300	0,1800	0,0250	0,0220	0,9100		0,1900												
L006	180002306	0,3900	0,7500	0,2200	0,0220	0,0210	1,0300		0,2000												
L007	218714	0,4100	0,8800	0,3400	0,0100	0,0020	1,0200		0,2310												
 29 MAG. 2023																					
OME Metallurgica Erbesse S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it							Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIOACTIVITY CONTAMINATION							Prepared  F. ROSSINI				Approved  M. MERONI			

INSPECTION CERTIFICATE					CERT. TYPE		CMTR Nr.		SHEET								
TENSILE TEST					ISO 10474 EN 10204		3.1 DATE 16/05/2023		3/6								
CUSTOMER : VALVITALIA S.P.A.					ORDER No.: OF000069504 REV.1 Date: 31/03/2023												
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY					PROJECT No.:												
					JOB No.:												
LOT	SPECIMEN				REQUIRED VALUES								OBTAINED VALUES				
	NO.	DIA. mm	AREA mm ²	L mm	T °C	Measure Unit	MIN Rp 0.2	R		MIN E %	MIN R.A. %	Rp 0.2	R	E %	R.A. %		
								MIN	MAX								
L003		12,5	122,66	50,00	Room	MPa	550	690		18	50	615	728	24,1	65,3		
L004		8,75	60,10	35,00	Room	MPa	550	690		18	50	620	723	23,0	64,0		
L007		12,5	122,66	50,00	Room	MPa	550	690		18	50	606	721	22,7	63,2		
OME Metallurgica Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it						Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION						Prepared 					

VALVITALIA S.P.A.
M.C.
29 MAG. 2023

INSPECTION CERTIFICATE						CERT. TYPE				CMTR Nr.				SHEET										
IMPACT TEST						ISO 10474 3.1 EN 10204				2300494/10 DATE 16/05/2023				4/6										
CUSTOMER : VALVITALIA S.P.A.						ORDER No.: OF000069504 REV.1 Date: 31/03/2023																		
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY						PROJECT No.:																		
						JOB No.:																		
LOT	SPECIMEN					MINIMUM REQUIRED VALUES					OBTAINED VALUES													
	NO.	IZOD	KV	ISOV	KCU	T °C	M.Unit	MIN.	AVE.	L.E.	ABSORBED ENERGY				LATERAL EXP.			SHEAR %						
											1	2	3	AVE.	1	2	3	60	70	80	90			
L001			X			-73	J	21	27		42	46	40	43										
L001			X			-73	J	21	27		43	44	44	44										
L002			X			-73	J	21	27		40	38	41	40										
L002			X			-73	J	21	27		43	42	39	41										
L003			X			-73	J	21	27		39	43	40	41										
L004			X			-73	J	21	27		41	41	45	42										
L005			X			-73	J	21	27		38	40	42	40										
L006			X			-73	J	21	27		44	42	38	41										
L007			X			-73	J	21	27		43	37	39	40										
 29 MAG. 2023																								
OME Metallurgica Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info@ome.it - http://www.ome.it						Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION						Prepared  F. ROSSINI						Approved  M. MERONI						
F 8.2 - CMTR - 06																								

INSPECTION CERTIFICATE										CERT. TYPE		CMTR Nr.		SHEET									
HARDNESS TEST										ISO 10474 3.1 EN 10204		2300494/10 DATE 16/05/2023		5/6									
CUSTOMER : VALVITALIA S.P.A.										ORDER No.: OF000069504 REV 1 Date: 31/03/2023													
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY										PROJECT No.:													
										JOB No.:													
LOT	NO. OF TEST	REQUIRED VALUES										OBTAINED VALUES											
		AFTER HEAT TREAT. x 24h				ROOM TEMPERATURE						AFTER HEAT TREAT. x 24h				ROOM TEMPERATURE							
		HBW	HRBW	T °C	HARD. MIN	HBW	HRBW	HRC	HV	HARDNESS		CONTR. QTY	HARDNESS		CONTR. QTY	HARDNESS		PL		CPL			
										MIN	MAX		MIN	MAX		MIN	MAX	C	NC	C	NC		
L001			X	590	84			X			22	2	91		2	19	21	X					
L001	*					X				159	235				100%	217	225						
L002			X	590	84			X			22	2	91		2	18	19	X					
L002	*					X				159	235				100%	210	220						
L003								X			22				1	19	20						
L003	*					X				200	235				100%	215	222						
L004								X			22				1	19	21						
L004	*					X				200	235				100%	216	228						
L005			X	590	84			X			22	1	89		1	18	20						
L005	*					X				159	235				100%	205	212						
L006			X	590	84			X			22	1	90		1	18	20						
L006	*					X				159	235				100%	213	224						
L007								X			22				1	17	19						
L007	*					X				200	235				100%	207	215						
 29 MAG. 2023																							
PL = Proof Load CPL = Cone Proof Load C = Conforming NC = Not Conforming																							
OME Metallurgica Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it										Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION										Prepared  F. ROSSINI		Approved  A. MERONI	
F 82 - CMTR - 07																							

INSPECTION CERTIFICATE		CERT. TYPE	CMTR Nr.	SHEET	
Declarations		ISO 10474 EN 10204 3.1	2300494/10 DATE 16/05/2023	6/6	
CUSTOMER : VALVITALIA S.P.A.		ORDER No.OF000069504 REV.1 Date: 31/03/2023:			
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY		PROJECT : No.:			
		JOB No.:			
Compliance ASTM A194-22 Ed API SPEC 20E 2nd Ed + A1.2018 + A2.2019 + E1.2021 MACROETCH ASTM E381; (1): S1; (2): R1; (3): C2; CONFORMING DECARBURIZATION TEST ASTM F2328 / F2328M Class 3: CONFORMING - Absence of gross and partial decarburization * HARDNESS TEST PERFORMED ON 100% OF PIECES ASTM A320-22 Ed ANSI/NACE MR-0175 / ISO 15156-2015 : CONFORMING			Material Specification VALVITALIA MDS_013 Rev 4 VALVITALIA MDS_012 Rev 4 VALVITALIA Annex A Rev 0		
Coating			Condition QTM = QUENCHED AND TEMPERED @ 1150 °F [620 °C] min Batch furnace - Furnace Calibration Acc to API 6A 21st Ed.		
Generic Specification API 20E BSL-1 7M (MPS 006) - A194 API 20E-1 : Identify thread forming process: Machining/cutting L7M (MPS 006) - A320 API 20E-1 : Identify thread forming process: Rolling					
OME Metallurgica Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it		Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION NO WELD REPAIR		Prepared  Approved  F 8.2 - CMTR - 30	


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INSPECTION CERTIFICATE				CERT. TYPE		CMTR Nr.	SHEET			
LIST OF SUPPLIED PRODUCTS				ISO 10474		2300494/7	1/3			
				EN 10204		DATE 05/05/2023				
CUSTOMER : VALVITALIA S.P.A.				ORDER No.: OF000069504 REV.1 Date: 31/03/2023						
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY				PROJECT No.:						
				JOB No.:						
LOT	OME ITEM	P.O. ITEM	Q. TY	Dimension	Drawing No.	Spec / Grade	Heat No.	Marking	Cond.	Coat.
L001	500	50	30 No.	3/4" 10 UNRC x 40	ASME B18.3-2012 TAB 1	ASTM A 320 / A 320M API 20E-1 L7M	3090786	STAMPINOX.IT- L7M-732	QTM	
L001	600	60	45 No.	3/4" 10 UNRC x 40	ASME B18.3-2012 TAB 1	ASTM A 320 / A 320M API 20E-1 L7M	3090786	STAMPINOX.IT- L7M-733	QTM	
L004	2600	260	60 No.	7/8" 9 UNRC x 47	ASME B18.3-2012 TAB 1	ASTM A 320 / A 320M API 20E-1 L7M	3090786	S-L7M-753	QTM	
L005	2700	270	30 No.	5/8" 11 UNRC x 40	ASME B18.3-2012 TAB 1	ASTM A 320 / A 320M API 20E-1 L7M	210553	STAMPINOX.IT- L7M-754	QTM	
L006	3200	320	30 No.	1" 8 UNRC x 67	ASME B18.3-2012 TAB 1	ASTM A 320 / A 320M API 20E-1 L7M	3090786	STAMPINOX.IT- L7M-759	QTM	
L004	3700	370	30 No.	7/8" 9 UNRC x 65	ASME B18.3-2012 TAB 1	ASTM A 320 / A 320M API 20E-1 L7M	3090786	STAMPINOX.IT- L7M-764	QTM	
L001	5100	510	40 No.	3/4" 10 UNRC x 50	ASME B18.3-2012 TAB 1	ASTM A 320 / A 320M API 20E-1 L7M	3090786	STAMPINOX.IT-L7 M-778	QTM	
 18 MAG. 2023										
VISUAL / DIMENSIONAL EXAMINATION : Conforming										
OME Metallurgica Erbesse S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it				Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION			Prepared  F. ROSSINI		Prepared  F. MEDICA F.	
							F 8.2 - CMTR - 20			

INSPECTION CERTIFICATE							CERT. TYPE			CMTR Nr.			SHEET					
HEAT ANALYSIS							ISO 10474 EN 10204			3.1 DATE 05/05/2023			2/3					
CUSTOMER : VALVITALIA S.P.A.							ORDER No.: OF000069504 REV.1 Date: 31/03/2023											
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY							PROJECT No.:											
							JOB No.:											
LOT	Heat No.	C	Mn	Si	P	S	Cr	Ni	Mo	V	Ti	Cu	W	Al	B	Nb	Co	
		Zn	Pb	Sn	Be	O	N	H	Fe	Nb+Ta	Cb	Al+Ti	Cb+Ta	Se	Bi	Ca	Mg	
L001	3090786	0,3900	0,8358	0,1977	0,0144	0,0050	0,9733		0,2154	0,0051		0,0926				0,0003		
L004	3090786	0,3900	0,8358	0,1977	0,0144	0,0050	0,9733		0,2154	0,0051		0,0926				0,0003		
L005	210553	0,4000	0,7800	0,2900	0,0120	0,0020	0,9700		0,2000	0,0030		0,1700				0,0004		
L006	3090786	0,3900	0,8358	0,1977	0,0144	0,0050	0,9733		0,2154	0,0051		0,0926				0,0003		
																		
OME Metallurgia Erbesse S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it							Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION											
F 8.2 - CMTR - 29																		

[illegible]

INSPECTION CERTIFICATE										CERT. TYPE					CMTR Nr.				SHEET					
IMPACT TEST										ISO 10474 3.1 EN 10204					2300494/7				3/3					
															DATE 05/05/2023									
CUSTOMER : VALVITALIA S.P.A.										ORDER No.: OF000069504 REV.1 Date: 31/03/2023														
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY										PROJECT No.:														
										JOB No.:														
LOT	SPECIMEN					MINIMUM REQUIRED VALUES					OBTAINED VALUES													
											ABSORBED ENERGY				LATERAL EXP.			SHEAR %						
	NO.	IZOD	KV	ISOV	KCU	T °C	M.Unit	MIN.	AVE.	L.E.	1	2	3	AVE.	1	2	3	60	70	80	90			
L001			X			-73	J	21	27		40	43	38	40										
L004			X			-73	J	21	27		46	41	41	43										
L005			X			-73	J	21	27		44	40	42	42										
L006			X			-73	J	21	27		43	46	40	43										
 18 MAG. 2023																								
OME Metallurgica Erbesse S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it										Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION										Prepared  F. ROSSINI		Prepared 		
F B 2 - CMTR - 06																								

INSPECTION CERTIFICATE										CERT. TYPE		CMTR Nr.		SHEET							
HARDNESS TEST										ISO 10474 3.1 EN 10204		2300494/7		3/3							
CUSTOMER : VALVITALIA S.P.A.										ORDER No.: OF000069504 REV.1 Date: 31/03/2023											
DESTINATION : VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY										PROJECT No.:											
										JOB No.:											
LOT	NO. OF TEST	REQUIRED VALUES										OBTAINED VALUES									
		AFTER HEAT TREAT. x 24h					ROOM TEMPERATURE					AFTER HEAT TREAT. x 24h					ROOM TEMPERATURE				
		HBW	HRBW	T °C	HARD. MIN	HBW	HRBW	HRC	HV	HARDNESS		CONTR. QTY	HARDNESS		CONTR. QTY	HARDNESS		PL		CPL	
										MIN	MAX		MIN	MAX		MIN	MAX	C	NC	C	NC
L001							X			22				1	18	20					
L001	*					X			200	235				100%	213	221					
L004							X			22				1	18	19					
L004	*					X			200	235				100%	210	217					
L005							X			22				1	19	20					
L005	*					X			200	235				100%	217	225					
L006							X			22				1	19	20					
L006	*					X			200	235				100%	215	225					


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PL = Proof Load		CPL = Cone Proof Load		C = Conforming		NC = Not Conforming	
OME Metallurgica Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it				Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION			
				F 8.2 - CMTR - 07			

INSPECTION CERTIFICATE		CERT. TYPE	CMTR Nr.	SHEET	
Declarations		ISO 10474 EN 10204	3.1 DATE 05/05/2023	3/3	
CUSTOMER : VALVITALIA S.P.A.		ORDER No. OF000069504 REV.1 Date: 31/03/2023:			
DESTINATION :		PROJECT : No.:			
VALVITALIA S.P.A. VIA DEL LAURO, 7 20121 - MILANO (MI) ITALY		JOB No.:			
Compliance ASTM A320-22 Ed. API 20E 2nd Ed + A1:2018 + A2:2019 + E1:2021 MACROETCH ASTM E381; (1): S1; (2): R1; (3): C2: CONFORMING DECARBURIZATION TEST ASTM F2328 / F2328M Class 3: CONFORMING - Absence of gross and partial decarburization * HARDNESS TEST PERFORMED ON 100% OF PIECES ANSI/NACE MR-0175 / ISO 15156-2015 : CONFORMING WEDGE TENSILE TEST ACCORDING TO ASTM A320: CONFORMING ANSI/NACE MR-0175 / ISO 15156-2015 : CONFORMING		Material Specification VALVITALIA MDS 012 Rev.4 VALVITALIA Annex A Rev.0			
Coating		Condition QTM = QUENCHED AND TEMPERED @ 1150 °F [620 °C] min Batch furnace – Furnace Calibration Acc.to API 6A 21st Ed.			
Generic Specification L7M (MPS 006) - A320 API 20E-1 : Identify thread forming process: Rolling					
OME Metallurgia Erbesi S.r.l. Via Milano, 15 22036 ERBA (CO) Italy Phone +39 031 641606 r.a. - Fax +39 031 645618 E-Mail: info.ome@ome.it - http://www.ome.it		Statement ALL THE MATERIALS SUPPLIED HAVE BEEN MANUFACTURED, EXAMINED AND TESTED SATISFACTORILY IN FULLY CONFORMITY WITH PURCHASE ORDER REQUIREMENTS AND APPLICABLE STANDARDS / SPECIFICATIONS. MATERIAL FREE FROM MERCURY OR RADIO-ACTIVITY CONTAMINATION NO WELD REPAIR		Prepared  Prepared  F 8.2 - CMTR - 30	



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		PRESSURE TEST CERTIFICATE Nr.P0485.23				Mod. 4.10.2 BVPT Rev. 6	
(Ref.Sper. API 6D, 25th. Ed)							
Customer:	QTR	VALVE: type	55	size:	10"	class:	900
Customer P.O.:	Q22-27052	Customer item:	4	VALVE SERIAL NUMBER:	R2217500040001		
VLT ODV:	R22175	VLT ITEM:	4	Actuator type/brand:	N.A.		
Test procedure No:	W.I.- 4.10.2	Rev.:	7	Actuator serial number:	N.A.		
TAG.:	N.A.	Gear :	YES	NO	☒		
TESTED SEQUENCE:		APPLIED/RECORDED VALUES				RESULT/NOTES	
ANTI-STATIC TESTING						Applicable: YES ☒ NO ☐	
Applied D.C. supply:		12 (V)					
Applied D.C. supply:		10 (Ω)		Measured resistance (Ω):		0,5	
HYDROSTATIC BODY TEST:		Body material: LF2					
Acceptance: No leakage allowed		Applied pressure (bar)		231			
		Test duration (min)		5		Satisfactory: YES ☒ NO ☐	
After stabilization.		Recorded leakage		N.L.			
2 nd HYDROSTATIC BODY TEST with external relief valve fitted		Applied pressure (bar)		180,5		Applicable: YES ☒ NO ☐	
		(95% of set pressure of the relief valve)					
Acceptance: No leakage allowed		Test duration (min)		5		Satisfactory: YES ☒ NO ☐	
		Recorded leakage		N.L.			
FUNCTIONAL TEST AT ATMOSPHERIC PRESSURE		three opening / closing operating cycles with the valve at atmospheric pressure				Satisfactory: YES ☒ NO ☐	
Acceptance: no sliding							
OPERATION TORQUE TEST		SIDE A		SIDE B		SIDE A+B	
Applied pressure (Δp):		153	bar	153	bar	153	bar
Max allowable BTO:		3.721	Nm	2405	Nm	2097	Nm
Max allowable running:		746	Nm	699	Nm	668	Nm
Max allowable ETC:		1.489	Nm	814	Nm	836	Nm
		SIDE A		SIDE B		BORE	
HYDROSTATIC SEAT TEST		Applied pressure (bar)		169		Satisfactory: YES ☒ NO ☐	
Max allowable leakage: RATE A: 0 ml/min		Test duration (min)		5			
		Recorded leakage		0 ml/min			
SELF RELIEVING TEST		SIDE A		SIDE B		Applicable: YES ☐ NO ☒	
Acceptance: max		bar		Relief pressure (bar)		Satisfactory: YES ☐ NO ☐	
HYDROSTATIC D.B.B. TEST		Applied pressure (bar)		169		Applicable: YES ☒ NO ☐	
		Test duration (min)		5			
Max allowable leakage: RATE A: 0 ml/min		Recorded leakage		0 ml/min		Satisfactory: YES ☒ NO ☐	
HYDROSTATIC D.P.E. TEST (DIB-1; DIB-2)		Applied pressure (bar)		169		Applicable: YES ☒ NO ☐	
		Test duration (min)		5			
Max allowable leakage: RATE A: 0 ml/min		Recorded leakage		0 ml/min		Satisfactory: YES ☒ NO ☐	
PNEUMATIC SEAT TEST Type I (1 Bar)		SIDE A		SIDE B		Applicable: YES ☐ NO ☒	
(AIR)		Applied pressure (bar)		Applied pressure (bar)			
Max allowable leakage:		Test duration (min)		Test duration (min)		Satisfactory: YES ☐ NO ☐	
After stabilization.		Recorded leakage		Recorded leakage			
PNEUMATIC SEAT TEST Type II (6 Bar)		SIDE A		SIDE B		Applicable: YES ☒ NO ☐	
(AIR)		Applied pressure (bar)		Applied pressure (bar)			
Max allowable leakage: RATE A: 0 cc/min		Test duration (min)		Test duration (min)		Satisfactory: YES ☒ NO ☐	
After stabilization.		Recorded leakage		Recorded leakage		0 cc/min	
HIGH PRESSURE GAS BODY TEST		Applied pressure (bar)				Applicable: YES ☐ NO ☒	
		Test duration (min)					
Max allowable leakage:		Recorded leakage				Satisfactory: YES ☐ NO ☐	
HIGH PRESSURE GAS SEAT TEST		SIDE A		SIDE B		Applicable: YES ☐ NO ☒	
		Applied pressure (bar)		Applied pressure (bar)			
Max allowable leakage:		Test duration (min)		Test duration (min)		Satisfactory: YES ☐ NO ☐	
		Recorded leakage		Recorded leakage			
FUNCTIONAL TEST WITH ACTUATOR		SIDE A		SIDE B		Applicable: YES ☐ NO ☒	
Applied pressure (Δp)		bar		bar			
Actuator Supply						Satisfactory: YES ☐ NO ☐	
Max allowable Opening time: (sec)		Opening time (sec)		Opening time (sec)			
Max allowable Closing time: (sec)		Closing time (sec)		Closing time (sec)			
TEST RESULT		OPERATOR NAME Andrea Depetris				DYNAMOMETRIC KEY ID	
Satisfactory ☒ internal ☒		TEST BENCH ID 4				NORBAR ID B 1721	
Not Satisfactory ☐ witness ☐		PRESSURE GAUGE(S) ID IT 106				OTHERS	
REMARKS:							
VALVITALIA SIGNATURE		TPI SIGNATURE			CUSTOMER SIGNATURE		
Andrea Depetris							
DATE: 13/06/2023		DATE:			DATE:		

VALVITALIA		PRESSURE TEST CERTIFICATE Nr.P0837.23				Mod. 4.10.2 BVPT Rev. 6	
(Ref.Sper. API 6D, 25th. Ed)							
Customer:	QTR	VALVE:	type	55	size:	10"	class: 900 end RF
Customer P.O.:	Q22-27426	Customer item:	4	VALVE SERIAL NUMBER: R2226600040001			
VLT ODV:	R22266	VLT ITEM:	4	Actuator type/brand: N.A.			
Test procedure No:	W.I.- 4.10.2	Rev.:	7	Actuator serial number: N.A.			
TAG.:	N.A.	Gear :	YES	NO	X		
TESTED SEQUENCE:		APPLIED/RECORDED VALUES				RESULT/NOTES	
ANTI-STATIC TESTING						Applicable: YES ☒ NO ☐	
Applied D.C. supply: 12 (V)							
Applied D.C. supply: 10 (V)		Measured resistance (Ω): 0,3				Satisfactory: YES ☒ NO ☐	
HYDROSTATIC BODY TEST:		Body material: LF2					
Acceptance: No leakage allowed		Applied pressure (bar) 231					
After stabilization.		Test duration (min) 5				Satisfactory: YES ☒ NO ☐	
		Recorded leakage No leakage					
2 nd HYDROSTATIC BODY TEST with external relief valve fitted		Applied pressure (bar) 180,5				Applicable: YES ☒ NO ☐	
Acceptance: No leakage allowed		(95% of set pressure of the relief valve)					
		Test duration (min) 5				Satisfactory: YES ☒ NO ☐	
		Recorded leakage No leakage					
FUNCTIONAL TEST AT ATMOSPHERIC PRESSURE		three opening / closing operating cycles with the valve at atmospheric pressure				Satisfactory: YES ☒ NO ☐	
Acceptance: no sliding							
OPERATION TORQUE TEST		SIDE A		SIDE B		SIDE A+B	
Applied pressure (Δp) 153		bar 153		bar 153		bar 153	
Max allowable BTO: 3721 Nm		Nm 2594		Nm 3146		Nm 3193	
Max allowable running: 746 Nm		Nm 614		Nm 585		Nm 546	
Max allowable ETC: 1489 Nm		Nm 779		Nm 648		Nm 618	
HYDROSTATIC SEAT TEST		SIDE A		SIDE B		BORE	
Applied pressure (bar) 169		bar 169		bar 169		bar 169	
Max allowable leakage: RATE A: 0ml/min		Test duration (min) 5		Test duration (min) 5		Test duration (min) 5	
		Recorded leakage 0 ml/min		Recorded leakage 0 ml/min		Recorded leakage 0 ml/min	
SELF RELIEVING TEST		SIDE A		SIDE B		Satisfactory: YES ☒ NO ☐	
Acceptance: max bar		Relief pressure (bar)		Relief pressure (bar)		Satisfactory: YES ☐ NO ☐	
HYDROSTATIC D.B.B. TEST		Applied pressure (bar) 169		Applied pressure (bar) 169		Applicable: YES ☒ NO ☐	
Max allowable leakage: RATE A: 0ml/min		Test duration (min) 5		Test duration (min) 5		Satisfactory: YES ☒ NO ☐	
		Recorded leakage 0 ml/min		Recorded leakage 0 ml/min			
HYDROSTATIC D.P.E. TEST (DIB-1; DIB-2)		Applied pressure (bar) 169		Applied pressure (bar) 169		Applicable: YES ☒ NO ☐	
Max allowable leakage: RATE A: 0ml/min		Test duration (min) 5		Test duration (min) 5		Satisfactory: YES ☒ NO ☐	
		Recorded leakage 0 ml/min		Recorded leakage 0 ml/min			
PNEUMATIC SEAT TEST Type I (1 Bar)		SIDE A		SIDE B		Applicable: YES ☐ NO ☒	
(AIR)		Applied pressure (bar)		Applied pressure (bar)		Satisfactory: YES ☐ NO ☐	
Max allowable leakage:		Test duration (min)		Test duration (min)			
After stabilization.		Recorded leakage		Recorded leakage			
PNEUMATIC SEAT TEST Type II (6 Bar)		SIDE A		SIDE B		Applicable: YES ☒ NO ☐	
(AIR)		Applied pressure (bar) 6		Applied pressure (bar) 6		Satisfactory: YES ☒ NO ☐	
Max allowable leakage: RATE A: 0cc/min		Test duration (min) 5		Test duration (min) 5			
After stabilization.		Recorded leakage 0 cc/min		Recorded leakage 0 cc/min			
HIGH PRESSURE GAS BODY TEST		Applied pressure (bar)		Applied pressure (bar)		Applicable: YES ☐ NO ☒	
Max allowable leakage:		Test duration (min)		Test duration (min)		Satisfactory: YES ☐ NO ☐	
		Recorded leakage		Recorded leakage			
HIGH PRESSURE GAS SEAT TEST		SIDE A		SIDE B		Applicable: YES ☐ NO ☒	
Max allowable leakage:		Applied pressure (bar)		Applied pressure (bar)		Satisfactory: YES ☐ NO ☐	
		Test duration (min)		Test duration (min)			
		Recorded leakage		Recorded leakage			
FUNCTIONAL TEST WITH ACTUATOR		SIDE A		SIDE B		Applicable: YES ☐ NO ☒	
Applied pressure (Δp)		bar		bar		Satisfactory: YES ☐ NO ☐	
Actuator Supply							
Max allowable Opening time: (sec)		Opening time (sec)		Opening time (sec)			
Max allowable Closing time: (sec)		Closing time (sec)		Closing time (sec)			
TEST RESULT		OPERATOR NAME Dellagiovanna		DYNAMOMETRIC KEY ID			
Satisfactory ☒ internal ☒		TEST BENCH ID 8		NORBAR ID B1360			
Not Satisfactory ☐ witness ☐		PRESSURE GAUGE(S) ID IT 832		OTHERS			
REMARKS:							
VALVITALIA SIGNATURE		TPI SIGNATURE		CUSTOMER SIGNATURE			
Dellagiovanna Alberto							
DATE: 30/08/2023		DATE:		DATE:			

		PRESSURE TEST CERTIFICATE Nr.P0837.23				Mod. 4.10.2 BVPT Rev. 6			
		(Ref.Sper. API 6D, 25th. Ed)							
Customer:	QTR	VALVE: type	55	size:	10"	class:	900	end	RF
Customer P.O.:	Q22-27426	Customer item:	4	VALVE SERIAL NUMBER:	R2226600040002				
VLT ODV:	R22266	VLT ITEM:	4	Actuator type/brand:	N.A.				
Test procedure No:	W.I.- 4.10.2	Rev.:	7	Actuator serial number:	N.A.				
TAG.:	N.A.	Gear :	YES	NO	☒				
TESTED SEQUENCE:		APPLIED/RECORDED VALUES				RESULT/NOTES			
ANTI-STATIC TESTING						Applicable: YES ☒ NO ☐			
Applied D.C. supply:		12 (V)							
Applied D.C. supply:		10 (Ω)		Measured resistance (Ω):		0,3		Satisfactory: YES ☒ NO ☐	
HYDROSTATIC BODY TEST:		Body material:				LF2			
Acceptance: No leakage allowed		Applied pressure (bar)		231					
		Test duration (min)		5		Satisfactory: YES ☒ NO ☐			
After stabilization.		Recorded leakage		No leakage					
2 nd HYDROSTATIC BODY TEST with external relief valve fitted		Applied pressure (bar)		180,5		Applicable: YES ☒ NO ☐			
		(95% of set pressure of the relief valve)							
Acceptance: No leakage allowed		Test duration (min)		5		Satisfactory: YES ☒ NO ☐			
		Recorded leakage		No leakage					
FUNCTIONAL TEST AT ATMOSPHERIC PRESSURE		three opening / closing operating cycles with the valve at atmospheric pressure				Satisfactory: YES ☒ NO ☐			
Acceptance: no sliding									
OPERATION TORQUE TEST		SIDE A		SIDE B		SIDE A+B		BORE	
Applied pressure (Δp):		153	bar	153	bar	153	bar	153	bar
Max allowable BTO:		3721	Nm	2983	Nm	3219	Nm	2460	Nm
Max allowable running:		746	Nm	730	Nm	673	Nm	649	Nm
Max allowable ETC:		1489	Nm	502	Nm	681	Nm	672	Nm
		684	Nm						
HYDROSTATIC SEAT TEST		SIDE A		SIDE B		SIDE A+B		BORE	
Applied pressure (bar)		169		169		169		169	
Max allowable leakage:		RATE A: 0ml/min		5		5		5	
		Recorded leakage		0 ml/min		0 ml/min		0 ml/min	
SELF RELIEVING TEST		SIDE A		SIDE B		SIDE A+B		BORE	
Acceptance: max		bar		bar		bar		bar	
HYDROSTATIC D.B.B. TEST		Applied pressure (bar)		169		Applied pressure (bar)		169	
		Test duration (min)		5		Test duration (min)		5	
Max allowable leakage:		RATE A: 0ml/min		0 ml/min		Recorded leakage		0 ml/min	
		Recorded leakage		0 ml/min		Recorded leakage		0 ml/min	
HYDROSTATIC D.P.E. TEST (DIB-1; DIB-2)		Applied pressure (bar)		169		Applied pressure (bar)		169	
		Test duration (min)		5		Test duration (min)		5	
Max allowable leakage:		RATE A: 0ml/min		0 ml/min		Recorded leakage		0 ml/min	
		Recorded leakage		0 ml/min		Recorded leakage		0 ml/min	
PNEUMATIC SEAT TEST Type I (1 Bar) (AIR)		SIDE A		SIDE B		SIDE A+B		BORE	
Applied pressure (bar)									
Max allowable leakage:									
After stabilization.		Recorded leakage		Recorded leakage		Recorded leakage		Recorded leakage	
PNEUMATIC SEAT TEST Type II (6 Bar) (AIR)		SIDE A		SIDE B		SIDE A+B		BORE	
Applied pressure (bar)		6		6		6		6	
Max allowable leakage:		RATE A: 0cc/min		5		5		5	
		Recorded leakage		0 cc/min		0 cc/min		0 cc/min	
After stabilization.		Recorded leakage		0 cc/min		Recorded leakage		0 cc/min	
HIGH PRESSURE GAS BODY TEST		Applied pressure (bar)				Applied pressure (bar)			
		Test duration (min)				Test duration (min)			
Max allowable leakage:		Recorded leakage				Recorded leakage			
HIGH PRESSURE GAS SEAT TEST		SIDE A		SIDE B		SIDE A+B		BORE	
Applied pressure (bar)									
Max allowable leakage:		Test duration (min)				Test duration (min)			
		Recorded leakage				Recorded leakage			
FUNCTIONAL TEST WITH ACTUATOR		SIDE A		SIDE B		SIDE A+B		BORE	
Applied pressure (Δp)		bar		bar		bar		bar	
Actuator Supply									
Max allowable Opening time: (sec)		Opening time (sec)		Opening time (sec)		Opening time (sec)		Opening time (sec)	
Max allowable Closing time: (sec)		Closing time (sec)		Closing time (sec)		Closing time (sec)		Closing time (sec)	
TEST RESULT		OPERATOR NAME		Dellagiovanna Alberto		DYNAMOMETRIC KEY ID			
Satisfactory ☒ internal ☒		TEST BENCH ID		8		NORBAR ID		B1360	
Not Satisfactory ☐ witness ☐		PRESSURE GAUGE(S) ID		IT 832		OTHERS			
REMARKS:									
VALVITALIA SIGNATURE		TPI SIGNATURE				CUSTOMER SIGNATURE			
<i>Dellagiovanna Alberto</i>									
DATE:		30/08/2023				DATE:			



BALL VALVE TEST RECORD

Date: 7/8/25WO#: Q34656-1

F-580-017, R6

CUST: Energy EquipmentP O #: 250504-2SPEC: Consumers 73700SIZE/CLASS/TYPE: 10" 900 Fig 55END CONNECTION: RFFEVALVE SER #: R221750004000/

TEST MEDIUM: ETHYLENE GLYCOL / WATER SOLUTION

ACTUATOR TYPE: —ACT. SER. #: —CHART ID #: 63GAGE ID #: 3649-4

HYDROSTATIC TEST

SHELL TEST: 3450 psi DURATION: 15 min.TORQUE @ FULL DIFF/MAOP: LEFT: — lb-ft RIGHT: — lb-ft MAX: — lb-ftDBB: — lb-ft DIB1: — lb-ftBREAKOUT @ Full DIFF/MAOP:
(Circle one)LEFT: PASS / FAIL (NA) RIGHT: PASS / FAIL (NA)DBB: PASS / FAIL (NA) DIB1: PASS / FAIL (NA)LEFT SEAT: 2445 psi DURATION: 5 minRIGHT SEAT: 2445 psi DURATION: 5 minDBB: 2445/2445 psi DURATION: 5/5 minDIB-1: 2445 psi DURATION: 5 minopen
closedAll Lube Lines Tested @ — psi for — min.

GAS / AIR TEST

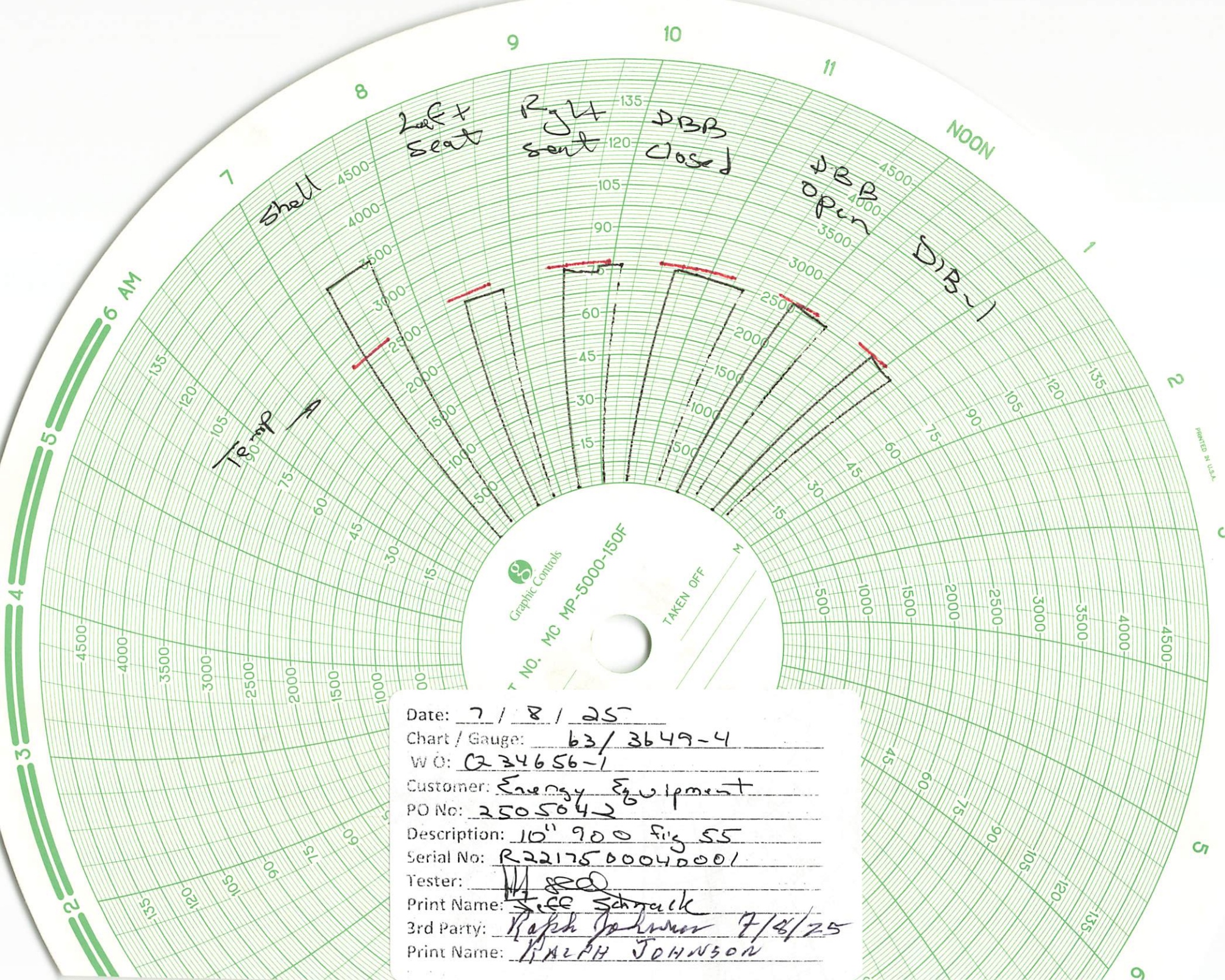
AIR SEAT LEFT: — psi DURATION: — min.AIR SEAT RIGHT: — psi DURATION: — min.DBB AIR: — psi DURATION: — min.DIB-1: — psi DURATION: — min.

By signature QTR certifies the above valve has satisfactorily passed all test indicated on this record.

-No visible leakage or inelastic deformation were evident during shell test

-No visible leakage evident during seat testing per ISO 5208 rate A

QTR [Signature] Date: 7/8/25Print: Jeff SchnackINSP (if Req'd): [Signature] Date: 7/8/25Print: RALPH JOHNSON



Date: 7 / 8 / 25

Chart / Gauge: 63 / 3649-4

W O: 0234656-1

Customer: Energy Equipment

PO No: 2505042

Description: 10" 900 Fig 55

Serial No: R2217500040001

Tester: M. R. R.

Print Name: Jeff Schuck

3rd Party: Ralph Johnson 7/8/25

Print Name: RALPH JOHNSON



BALL VALVE TEST RECORD

Date: 7/8/25WO#: Q34656-1

F-580-017, R6

CUST: Energy EquipmentP O #: 250504-2SPEC: Consumers 73700SIZE/CLASS/TYPE: 10" 900 Fig 55END CONNECTION: RFFEVALVE SER #: R222660004000 /

TEST MEDIUM: ETHYLENE GLYCOL / WATER SOLUTION

ACTUATOR TYPE: —ACT. SER. #: —CHART ID #: 62GAGE ID #: 6993-1

HYDROSTATIC TEST

SHELL TEST: 3450 psi DURATION: 15 min.TORQUE @ FULL DIFF/MAOP: LEFT: — lb-ft RIGHT: — lb-ft MAX: — lb-ftDBB: — lb-ft DIB1: — lb-ftBREAKOUT @ Full DIFF/MAOP:
(Circle one)LEFT: PASS / FAIL (NA) RIGHT: PASS / FAIL (NA)DBB: PASS / FAIL (NA) DIB1: PASS / FAIL (NA)LEFT SEAT: 2445 psi DURATION: 5 minRIGHT SEAT: 2445 psi DURATION: 5 minDBB: 2445/2445 psi DURATION: 5/5 minDIB-1: 2445 psi DURATION: 5 minopen
closedAll Lube Lines Tested @ — psi for — min.

GAS / AIR TEST

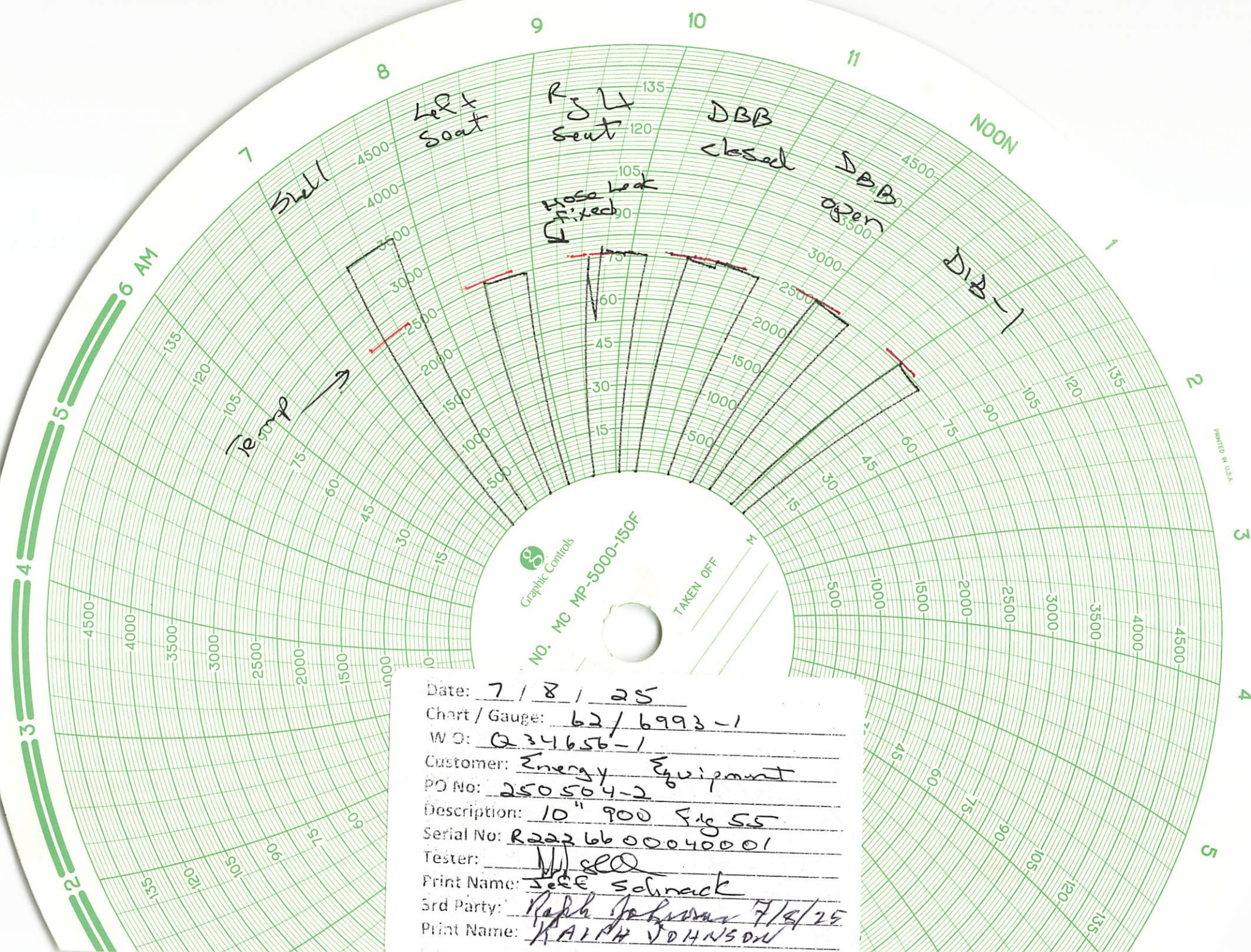
AIR SEAT LEFT: — psi DURATION: — min.AIR SEAT RIGHT: — psi DURATION: — min.DBB AIR: — psi DURATION: — min.DIB-1: — psi DURATION: — min.

By signature QTR certifies the above valve has satisfactorily passed all test indicated on this record.

-No visible leakage or inelastic deformation were evident during shell test

-No visible leakage evident during seat testing per ISO 5208 rate A

QTR [Signature] Date: 7/8/25Print: Jeff SchnackINSP (if Req'd): [Signature] Date: 7/8/25Print: RALPH JOHNSON



Date: 7 / 8 / 25
Chart / Gauge: 62 / 6993-1
W O: Q 34656-1
Customer: Energy Equipment
PO No: 250504-2
Description: 10" 900 Fig 55
Serial No: R2226600040001
Tester: MJL
Print Name: Jeff Schnack
3rd Party: Ralph Johnson 7/8/25
Print Name: RAIMON JOHNSON



BALL VALVE TEST RECORD

Date: 7/8/25

WO#: Q34656-1

F-580-017, R6

CUST: Energy Equipment

P O #: 250504-2

SPEC: Consumers 73700

SIZE/CLASS/TYPE: 10" 900 Fig 55

END CONNECTION: RFFE

VALVE SER #: R2226600040002

TEST MEDIUM: ETHYLENE GLYCOL / WATER SOLUTION

ACTUATOR TYPE: —

ACT. SER. #: —

CHART ID #: 64

GAGE ID #: 3649-2

HYDROSTATIC TEST

SHELL TEST: 3450 psi

DURATION: 15 min.

TORQUE @ FULL DIFF/MAOP:

LEFT: — lb-ft

RIGHT: — lb-ft

MAX: — lb-ft

DBB: — lb-ft

DIB1: — lb-ft

BREAKOUT @ Full DIFF/MAOP:
(Circle one)

LEFT: PASS / FAIL (NA)

RIGHT: PASS / FAIL (NA)

DBB: PASS / FAIL (NA)

DIB1: PASS / FAIL (NA)

LEFT SEAT: 2445 psi

DURATION: 5 min

RIGHT SEAT: 2445 psi

DURATION: 5 min

DBB: 2445/2445 psi

DURATION: 5/5 min

DIB-1: 2445 psi

DURATION: 5 min

open
closed

All Lube Lines Tested @ — psi for — min.

GAS / AIR TEST

AIR SEAT LEFT: — psi

DURATION: — min.

AIR SEAT RIGHT: — psi

DURATION: — min.

DBB AIR: — psi

DURATION: — min.

DIB-1: — psi

DURATION: — min

By signature QTR certifies the above valve has satisfactorily passed all test indicated on this record.

-No visible leakage or inelastic deformation were evident during shell test

-No visible leakage evident during seat testing per ISO 5208 rate A

QTR

Date: 7/8/25

Print: Jeff Schnack

INSP (if Req'd):

Date: 7/8/25

Print: Ralph Johnson



Date: 7/8/25
Chart / Gauge: 64/3649-2
W.O.: Q34656-1
Customer: Energy Equipment
P.O. No.: 250504-2
Description: 10" 900 Fig 55
Serial No: R2226600040002
Tester: M. Reed
Print Name: Jeff Schnack
3rd Party: Ralph Johnson 7/6/25
Print Name: KALAN JOHNSON

Customer:	Energy Equipment	PO#	250504-2
Q#	34656-1	WI#	136-F-003 (Consumers Green)
QTY#	3 Description: 10" 900# RFFE Fig 55 Delta Ball Valve, KE40 Gear w/HW		

Surface Preparation

Surface free of Contamination:	Yes	Solvent Cleaned:	SSPC-SP-1
Soluble Salt test:	Yes	Salt test performed:	Potassium Ferricyanide
Blast Pressure:	110 psi	Test Plate:	Yes
Blotter Test:	ASTM D 4285	Abrasive Type:	Medium
Surface Profile Standard:	SSPC-SP-10	Blast Acceptance:	SP-10
Profile required	2-4 mils	Profile Average:	2.93
Testex Tape Sheet Attached:	NA	Blast time:	7-10-25 8:15am

Blast profile Tape



Coating Conditions

Coat	RH%	TA°	TS°	TD°	TsTd°	Date	Time
1st	72.4	86.5	84.1	76.6	7.5	7/10/2025	9:05am
2nd	61.9	85.4	88.2	70.8	17.3	7/11/2025	8:45am
3rd							

Coating Materials

COAT	MANUFACTURER	PAINT	COLOR	BATCH A#	BATCH B#	Temp
1st	Sherwin Williams	Dura-Plate	White	XM3034XJ	XM3534NZ	76°
2nd	Sherwin Williams	Hi-Solids Poly	Consumers Grn	XM0455NU	MQ3584JW	76°
3rd						

Coating DFT

SN#	Stripe	1st Coat	2nd Coat	3rd Coat	4th Coat	Total
R2217500040001		9.72	14.08			14.08
R2226600040001		9.54	14.3			14.3
R2226600040002		9.27	13.55			13.55

Q34656-1

FINAL INSPECTION

Customer Specification #

Consumers 73700

RESULTS

DFT Avg.	Gauge:	SN#	Cert Date:
13.97	PosiTector	232259	10/23/2025

HOLIDAY TEST

Conducted:	High Voltage	Low Voltage	Voltage
NA			

Gauge:		SN#			Cert Date:

Comments:

Apply 2-part consumers green to a final dft range of 10-16mils min.

Exp. Date:

Dura-Plate 235 A. 10/29/2027 B. 12/18/2026

Hi-Solids Poly S. 2/14/2028 T. 12/23/2026

QTR Inspector: J. Brayer Nace Cert# 72547 Date: 7-15-25Customer Inspector: Roger John Nace Cert# 110552 Date: 7-16-25